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Unsupervised Face Embedding Clustering for Model Identity Recognition

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Darmstadt, August 18, 2025

Ritvik Maini

ABSTRACT

Identifying when the same face appears across very large image libraries is a core machine-learning challenge: labels are scarce, privacy matters and the data are messy. This thesis tackles that problem with a fully unsupervised pipeline that (i) detects faces, (ii) turns each face into a compact numeric fingerprint ("embedding"), (iii) lightly compresses those embeddings and (iv) groups them into identity-coherent clusters. The study spans a real, catalog-scale corpus of over 0.45 million images, aiming for methods that are not just accurate on paper but trustworthy and practical.

The central result is deliberately simple: ArcFace + PCA + DBSCAN consistently forms tight, credible identity clusters without any labels, and safely leaves ambiguous images as noise rather than forcing bad merges. By contrast, pairing UMAP with a graph label-propagation method (Chinese Whispers) can produce dazzling internal scores while over-segmenting identities into many tiny groups, useful in some settings but easy to misread if those scores are taken at face value. Taken together, the findings offer a clear recipe: stabilize the geometry (PCA) and prefer conservative, density-based grouping (DBSCAN) when trustworthiness matters most.

Beyond methodological insight, the work shows how unsupervised face clustering can support real-world needs, organizing vast image archives, enabling privacy-conscious search and laying groundwork for human-in-the-loop review, while being honest about limits (no ground-truth labels, look-alikes, occlusions) and the path to closing them with incremental assignment and light supervision.

ZUSAMMENFASSUNG

Das Erkennen, wann dasselbe Gesicht in sehr großen Bildsammlungen vorkommt, ist eine zentrale Herausforderung des maschinellen Lernens: Beschriftungen sind selten, Datenschutz ist wichtig und die Daten sind uneinheitlich. Diese Arbeit adressiert das Problem mit einer vollständig unüberwachten Pipeline, die Gesichter erkennt, jedes Gesicht in eine kompakte numerische Signatur (512-dimensionale ArcFace-Einbettung) überführt, diese Signaturen moderat komprimiert und anschließend zu identitätskohärenten Gruppen zusammenfasst. Die Untersuchung stützt sich auf einen realen Korpus mit deutlich mehr als 450 000 Bildern und verfolgt Methoden, die nicht nur auf dem Papier überzeugen, sondern sich auch als vertrauenswürdig und praktisch erweisen.

Das zentrale Ergebnis ist bewusst schlicht: ArcFace mit PCA und anschließendem DBSCAN bildet ohne Labels konsistent enge und belastbare Cluster und lässt unklare Fälle sinnvoll als Rauschen zurück, statt fehlerhafte Zusammenschlüsse zu erzwingen. Demgegenüber kann die Kopplung von UMAP mit einem graphbasierten Label-Propagation-Verfahren (Chinese Whispers) beeindruckende interne Kennzahlen liefern, neigt jedoch dazu, Identitäten in viele Kleinstgruppen zu zerlegen – in manchen Situationen nützlich, aber leicht fehlzuinterpretieren, wenn man diese Werte unkritisch übernimmt. Zusammengenommen ergibt sich ein klares Rezept: die Geometrie durch PCA stabilisieren und bei hohem Vertrauensbedarf konservative, dichtebasierte Gruppierungen mit DBSCAN bevorzugen.

Über die Methodik hinaus zeigt die Arbeit, wie unüberwachtes Gesichts-Clustering praktische Aufgaben unterstützt, etwa die Strukturierung großer Bildarchive und datenschutzbewusste Suche, und sie benennt offen die Grenzen wie fehlende Ground-Truth-Labels, Doppelgänger und Okklusionen sowie den Weg von der Theorie in den Einsatz über inkrementelle Zuordnung und maßvolle menschliche Rückmeldung.

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ABBREVIATIONS

ANN Approximate Nearest Neighbour

API Application Programming Interface

ARI Adjusted Rand Index

ARO Approximate Rank-Order

CFP-FP Celebrities in Frontal-Profile (Face Recognition Benchmark)

CH Calinski–Harabasz Index

CNN Convolutional Neural Network

CW Chinese Whispers

DBCV Density-Based Clustering Validation

DBI Davies–Bouldin Index

DBSCAN Density-Based Spatial Clustering of Applications with Noise

DRY Don't Repeat Yourself

EOM Excess-of-Mass

FAISS Facebook AI Similarity Search

GDPR General Data Protection Regulation

GPU Graphics Processing Unit

HDBSCAN Hierarchical DBSCAN

IQR Inter-Quartile Range

KNN k-Nearest Neighbours

L2 Euclidean *L2* norm

LFW Labeled Faces in the Wild (Face Recognition Benchmark)

MAP Mean Average Precision

MST Minimum-Spanning Tree

MTCNN Multi-Task Cascaded Convolutional Networks

ONNX Open Neural Network Exchange

PCA Principal Component Analysis

RAM Random Access Memory

SKU Stock Keeping Unit

SVD Singular Value Decomposition

UML Unified Modeling Language

UMAP Uniform Manifold Approximation and Projection

YOLO You Only Look Once

Part I

THESIS

1 INTRODUCTION

This thesis explores how unsupervised face-identity clustering can transform the management of large-scale model photography in fashion e-commerce, delivering operational efficiency and GDPR-compliant control of visual assets.

1.1 ACADEMIC CONTEXT AND STATE OF THE ART

Over the last decade deep learning has progressed from an academic curiosity to a work-horse technology. Within that shift, *computer vision* now enables machines to understand images at a scale and accuracy that once seemed impossible, unlocking new efficiencies in fields as diverse as autonomous driving, medical diagnostics, and—central to this thesis—digital-asset management for online retail.

Modern identity-related vision systems rely on *deep metric learning*. Instead of classifying images directly, a network learns an embedding space in which geometric distance reflects semantic similarity. Landmark studies such as FaceNet [SKP15] and ArcFace [Den+19a] show that a simple cosine distance on 512-dimensional vectors can discriminate faces with near-human accuracy, even under changes in pose, lighting, or expression. These pre-trained models form the first technical pillar of this work.

Yet a high-quality embedding alone does not solve an industrial problem when no ground-truth labels exist. The second pillar is *unsupervised clustering*: algorithms that discover natural groupings without prior annotations. Choosing an algorithm—density-based, hierarchical, or graph-based—is data-dependent and remains an open research question. The science at the heart of this thesis therefore asks: *How can rich, high-dimensional face embeddings be partitioned effectively, at scale, and without labels, to address a real business need?*

1.2 MOTIVATION AND RELEVANCE

The rapid growth of online fashion retail has turned product imagery from a marketing asset into an operational backbone. Companies such as the thesis partner, **TB International GmbH**, maintain catalogues containing 455 948 photographs across 74 152 stock-keeping units. Knowing which professional model appears in which image is critical: it supports copyright tracking, ensures a coherent brand presence, and helps reduce return rates by maintaining consistent visuals [Ade+21]. Regulations such as the GDPR further demand tight control over personal data, including model portraits [Eur16].

Most retailers, however, still rely on manual or fragmented tagging workflows—slow, error-prone, and unscalable. Automating identity discovery therefore promises immediate value: fewer staff hours, richer analytics, and a clear compliance trail.

1.3 PROBLEM STATEMENT AND OBJECTIVES

This thesis tackles the automatic grouping of unlabelled model photographs by facial identity. Three hurdles define the task:

(a) large variation in resolution, pose, and styling;

(b) complete absence of ground-truth identity labels;

(c) data volume in the hundreds of thousands of images.

The overarching objective is to *design and evaluate a pipeline that detects faces, extracts numerical embeddings, and clusters those embeddings into coherent identity groups*. Evaluation focuses on cluster compactness, coverage, and the feasibility of integrating the pipeline into TB International's production environment.

1.4 PROPOSED APPROACH

The solution unfolds in four stages:

(i) **Pose-aware curation** — select the most face-friendly shot for each article and standardise resolution.

(ii) **Embedding extraction** — run three independent detector–recogniser stacks (InsightFace, YOLOv8 + ArcFace, DeepFace) to obtain 512-D vectors.

(iii) **Dimensionality reduction** — apply PCA and UMAP to study the trade-off between speed and representational fidelity.

(iv) **Unsupervised clustering** — compare density-based, variable-density, hierarchical, and graph-based algorithms on real-world data.

By systematically varying parameters across all stages, the thesis identifies operating points that balance precision, recall, and runtime—results that can be deployed directly within TB International's asset workflow and generalised to other fashion retailers.

2 BACKGROUND AND RELATED WORK

This chapter introduces the technical foundations of face recognition and face embeddings, outlines clustering methods for grouping identities without labels, and situates our approach within prior work. The goal is to bridge the problem motivation from the introduction with the implementation details developed in the methodology.

2.1 FUNDAMENTALS OF FACE RECOGNITION AND EMBEDDINGS

Modern face recognition systems follow a common pipeline with four stages: *detection*, *alignment*, *feature extraction*, and *matching/classification*. Detection localizes faces; alignment normalizes pose and scale using landmarks; feature extraction maps an aligned crop to a compact numerical representation (the *embedding*); and matching uses distances or similarities between embeddings for verification or identification. In this thesis we focus on detection/alignment and embedding, as these stages determine the quality of the representation used for unsupervised clustering.

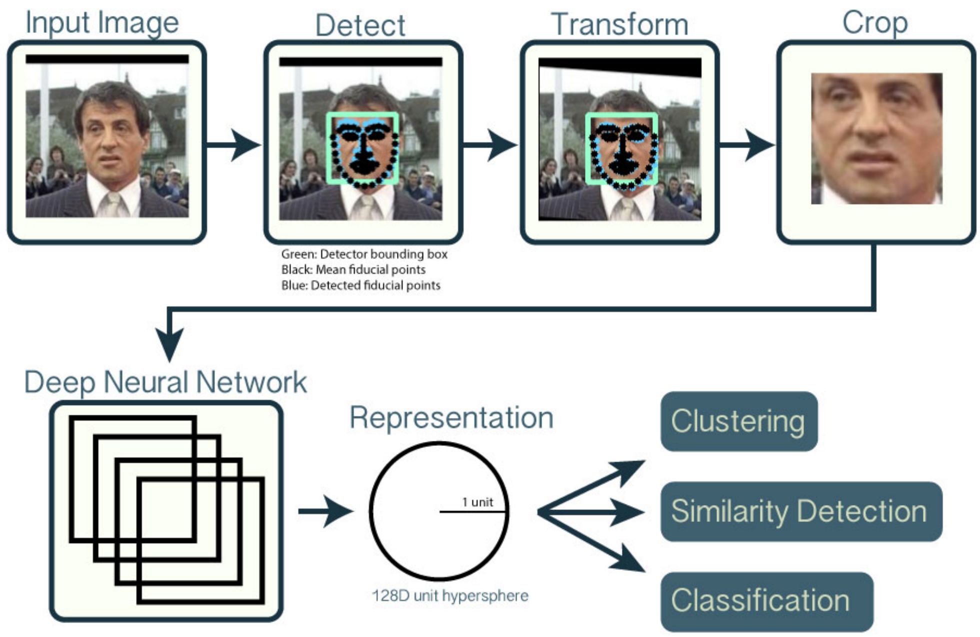

Figure 2.1: Canonical face recognition pipeline from detection to matching. Based on OpenFace [ALS16].

2.1.1 *Face detection and embeddings*

DETECTION AND ALIGNMENT. Early face detectors relied on boosted Haar-like features [VJ01]. Current practice uses deep detectors that are more robust to variation in pose, scale, and illumination. MTCNN couples detection and alignment via a cascade of CNNs and landmark regression [Zha+16]. RetinaFace advances this further with single-shot detection and facial landmark prediction, delivering strong results on challenging benchmarks [Den+20]. For high-throughput settings, one-stage object detectors such as YOLO adapt well to face detection, trading some localization accuracy for speed (the original YOLO reports real-time performance [Red+16]). In this work we adopt RetinaFace for accuracy and also consider a YOLO-based detector for throughput; implementation specifics are detailed in the methodology. Regardless of detector, aligned crops (e.g., by rotating and scaling using eye landmarks) reduce intra-class variation and are known to improve recognition accuracy [Tai+14].

EMBEDDINGS. A face embedding is a vector $\mathbf{f} \in \mathbb{R}^d$ that encodes identity information so that geometric proximity corresponds to visual similarity. FaceNet popularized training with a metric-learning objective to map faces to a compact space, achieving strong verification accuracy on LFW [SKP15; Hua+07]. Subsequent models refine the embedding geometry. ArcFace introduces an additive angular margin on a unit hypersphere, enforcing tighter intra-class compactness and larger inter-class separation [Den+19a], and has reported state-of-the-art results relative to earlier margin-based losses such as SphereFace and CosFace [Liu+17; Wan+18]. In practice, embeddings are L_2-normalized so that cosine similarity (angular distance) becomes the natural comparison metric [Wan+17].

High-dimensional embeddings (e.g., 512-D) are expressive but can challenge distance-based grouping due to concentration effects in high dimensions [Bey+99; AHK01]. Consistent normalization mitigates some issues, and later we consider dimensionality reduction (e.g., PCA) as a lightweight way to improve separability for clustering while deferring algorithmic details to the methodology. Overall, the reliability of modern hyperspherical embeddings underpins our unsupervised strategy: if nearby vectors correspond to the same identity, clustering by proximity becomes a viable approach.

2.1.2 *Face Embedding Models in This Work*

This thesis employs multiple pipelines that pair a face detector with an embedding network. We briefly introduce each configuration and explain why it is included in our evaluation.

INSIGHTFACE (ARCFACE). InsightFace is an open-source toolkit that provides high-performing models for detection, alignment, and recognition [DG+18]. We use its ArcFace-based recognizer (e.g., the `buffalo_l` variant) as a primary embedding generator. Given an aligned face crop, the network outputs a 512-dimensional, L_2-normalized vector suitable for cosine-similarity comparisons [Den+19a]. For detection and landmarks, InsightFace supports strong single-shot detectors such as RetinaFace and efficient variants like SCRFD [Den+20; Guo+21]. This pairing offers accurate localization and consistent alignment, which reduces intra-class variation before embedding [Tai+14]. We adopt this pipeline as a high-accuracy baseline and to reflect widely used practice in industry and research.

YOLOV8-FACE AND ARCFACE. To explore a higher-throughput alternative, we integrate a YOLO-based face detector (a YOLOv8 model fine-tuned for faces) with an ArcFace embedding network. YOLOv8 follows the modern single-stage design that emphasizes real-time inference, making it attractive for large-scale processing [Joc+23]. After detection and alignment, we compute 512-D ArcFace embeddings (exported in ONNX for efficient runtime) to maintain comparability with the InsightFace setup [Den+19a]. This configuration intentionally trades some detection robustness for speed—an expected trade-off for single-stage detectors—allowing us to quantify throughput/accuracy effects at catalog scale [Ge+21].

DEEPFACE (RETINAFACE AND ARCFACE). We also evaluate the DeepFace Python library as an off-the-shelf pipeline [Ser20]. DeepFace provides a uniform interface to multiple detectors and backbones; here we select RetinaFace for detection and ArcFace for recognition, mirroring the components above via an independent implementation [Den+20; Den+19a]. DeepFace returns 512-D embeddings that are L_2-normalized by default, which facilitates direct comparison across toolchains. Using this third implementation guards against tool-specific effects and improves the robustness of our conclusions.

SUMMARY AND RATIONALE. Across all configurations, the output representation is a 512-D, L_2-normalized embedding designed for angular (cosine) similarity on a hypersphere [Den+19a; Wan+17]. Using multiple but architecturally compatible pipelines allows us to separate the influence of detection/alignment from the embedding model and to assess the speed–accuracy trade-offs relevant to TB International's large-scale setting. Because our task is *unsupervised identity grouping*, these representations are the core signal: if embeddings of the same person are close in the feature space, clustering by proximity should recover identities without labels.

2.2 UNSUPERVISED CLUSTERING ALGORITHMS

Unsupervised clustering is central to grouping face embeddings by identity when ground-truth labels are unavailable and the number of identities is unknown. By operating directly on feature vectors learned by face-recognition models, clustering can discover sets of images that correspond to the same person based on proximity in the embedding space. This is particularly appropriate at TB International's scale, where manual labeling is infeasible and identities are not fixed in advance. In our setting—fashion model identification from unlabeled catalog images—clustering serves to aggregate near-duplicate or related shots of the same model into coherent identity groups, effectively yielding pseudo-labels for downstream analysis. Prior work has noted that face clustering in unconstrained conditions (pose change, lighting variation, occlusion) remains challenging and relatively under-explored compared to supervised recognition, underscoring the need for robust methods that tolerate real-world variability [Roy+20; Yan+20].

We focus on density-based, hierarchical, and graph-based approaches (DBSCAN, HDBSCAN, agglomerative hierarchical clustering, and Chinese Whispers) because they better align with the constraints of this problem than partitioning methods. In particular, unlike k-means, these methods do not require the number of clusters to be specified a priori—an impractical requirement when the true number of identities is unknown and variable [Llo82]. Spectral clustering can produce strong partitions, but its reliance on building and decomposing an $n \times n$ affinity matrix limits scalability (eigendecomposition is often cubic in n in practice) and introduces additional graph-construction hyperparameters, while offering no native mechanism for outlier handling [NJW02; Lux07]. In contrast, the methods we adopt can discover an appropriate number of clusters from data and explicitly label low-density observations as noise—both desirable properties for large, noisy embedding sets. Finally, supervised or semi-supervised variants that rely on labels or pairwise constraints fall outside our scope: we assume no identity labels, no must-/cannot-link cues, and treat the task as fully unsupervised [Roy+20].

2.2.1 *Density-Based Methods: DBSCAN and HDBSCAN*

DBSCAN. DBSCAN (Density-Based Spatial Clustering of Applications with Noise) defines clusters as contiguous regions of high density separated by low-density areas [Est+96a]. Points with at least a minimum number of neighbors within a radius form *core points*, while nearby non-core points are *border points*; points not density-reachable are labeled as noise. Key advantages include: (i) automatic estimation of the number of clusters, (ii) discovery of non-convex cluster shapes, and (iii) explicit outlier detection. With spatial indexing, practical runtimes are near $O(n \log n)$ on many datasets, supporting large-scale use [Sch+17]. In face-embedding applications, DBSCAN's ability to separate dense identity regions while discarding ambiguous samples helps preserve cluster purity in the absence of labels.

LIMITATIONS. DBSCAN employs a global density threshold, which can be brittle when clusters have heterogeneous densities; a single choice of neighborhood radius may fragment dense identities or merge sparser ones. Performance can also degrade in high dimensions as distances concentrate, making density estimation less reliable and motivating the use of normalization and (optionally) dimensionality reduction prior to clustering [Bey+99; AHK01].

HDBSCAN. HDBSCAN (Hierarchical DBSCAN) addresses the fixed-threshold limitation by building a hierarchy over varying density levels and selecting clusters according to a *stability* criterion rather than a fixed radius [CMS13a; Cam+15]. In practice, users specify only a minimum cluster size; the method then adapts to variable-density structure, retains DBSCAN's noise labeling, and removes the need for sensitive radius tuning. This adaptability is valuable for facial datasets where some identities appear in many images (high-density regions) while others are sparse.

TRADE-OFFS. HDBSCAN incurs higher computational overhead due to graph construction and hierarchy extraction (often approaching quadratic behavior in worst cases), though optimized implementations handle large datasets effectively. Its stability preference may label borderline groups as noise, which can be beneficial for purity but may reduce recall in edge cases [Cam+15].

RATIONALE FOR INCLUSION. DBSCAN provides a transparent baseline with intuitive density control, while HDBSCAN extends robustness to variable densities with less parameter sensitivity. Together, they satisfy our requirements for automatic cluster discovery, outlier handling, and scalability, and they align with the practical demands of clustering millions of high-dimensional face embeddings in an unlabeled setting.

2.2.2 *Hierarchical Methods: Agglomerative Clustering*

Hierarchical clustering is a classic family of methods dating back to early data analysis literature. In its agglomerative (bottom-up) form, each sample starts as its own cluster and the algorithm repeatedly merges the most similar pair until a stopping rule is met. The resulting dendrogram encodes cluster structure across scales—from fine partitions to a single merge at the root—and allows practitioners to select a resolution by "cutting" the tree at an appropriate level [JMF99]. Merge decisions are controlled by the linkage criterion: single linkage uses nearest neighbors, complete linkage uses farthest neighbors, average linkage aggregates pairwise distances, and Ward's method minimizes within-cluster variance across merges.

CORE PRINCIPLES AND PARAMETERS. Agglomerative clustering requires (i) a similarity or distance between points (for face embeddings, cosine distance on L_2-normalized vectors is standard; Euclidean is also common) and (ii) a linkage rule to compute inter-cluster distances. Iterative merges produce a dendrogram with leaves as data points and internal nodes as merges [Mül11]. Because the dendrogram is a multiscale representation, cluster count need not be specified a priori; a threshold on tree height defines the partition. While there is no explicit noise model, stringent cuts often isolate tiny or singleton clusters that effectively act as outliers, a practice used in recent face-clustering work [Hu+21].

USE CASES AND STRENGTHS. Hierarchical clustering is widely used in genomics, document analysis, and vision. In face analysis, early work exploited single-linkage to group temporally adjacent frames; later applications favored average/complete linkage to avoid the "chaining" effect of single linkage and to produce more compact, interpretable groups [JMF99; Mül11]. A major advantage is interpretability: the dendrogram reveals how clusters form and split, helping diagnose ambiguous identities and supporting principled threshold selection when clear gaps exist.

LIMITATIONS. Classical agglomerative algorithms can be computationally heavy; worst-case complexity approaches $O(n^3)$, though optimized implementations reduce this substantially and are practical for large n in many settings [Mül11]. The greedy nature of merges is irreversible—early errors cannot be undone—and the method lacks explicit noise handling, so outliers may be absorbed into nearby clusters. Performance depends strongly on the quality of the metric; poor similarity measures degrade results. In this thesis, robust hyperspherical embeddings (ArcFace) and cosine distance mitigate these risks by supplying a semantically meaningful geometry for clustering.

RATIONALE FOR METHOD SELECTION. Agglomerative clustering is included for its interpretability and flexibility: thresholds over the dendrogram infer the number of identities without supervision and can be tuned for purity or coverage. The hierarchical view provides complementary evidence alongside density-based methods, especially when we need insight into borderline merges and relative cluster similarities [JMF99].

2.2.3 *Graph-Based Methods: Chinese Whispers*

Chinese Whispers (CW) is a fast, label-propagation algorithm on undirected weighted graphs originally proposed for NLP tasks [Bie06a]. Nodes represent samples and edges encode pairwise affinities (e.g., a k-nearest-neighbor graph in embedding space). Initialization assigns a unique label to each node; the algorithm then iteratively relabels nodes to the most frequent label among their neighbors, propagating community structure until convergence. Despite the randomized update order, CW typically converges quickly and runs in time linear in the number of edges, making it attractive for large-scale problems.

CORE PRINCIPLES AND PARAMETERS. CW assumes a precomputed similarity graph. In practice, graph construction (choice of k, similarity threshold, and edge weights) sets the effective clustering scale: denser graphs yield fewer, larger communities; sparser graphs produce more, smaller ones. The algorithm itself has minimal hyperparameters and terminates when labels stop changing. Isolated nodes naturally emerge as singletons, which can be interpreted as outliers [CWL19].

USE CASES AND STRENGTHS. Beyond language tasks, CW has been applied to image and face clustering due to its simplicity, speed, and ability to capture arbitrarily shaped communities via connectivity [Bie06a; CWL19]. On large embedding collections, the linear-in-edges complexity makes CW a practical alternative to methods that require building dense affinity matrices, and its community-detection flavor aligns well with the notion of identity groups connected through local neighbors.

LIMITATIONS AND TRADE-OFFS. CW's outcomes depend heavily on graph construction; overly sparse graphs can fragment identities into many small clusters, and tie-breaking in local updates can introduce variability [CWL19]. Because CW lacks an explicit distance threshold, fine control over cluster boundaries is delegated to the graph (choice of k, weighting, pruning). In practice, light post-processing—such as merging highly similar small components—can reduce over-segmentation.

RATIONALE FOR METHOD SELECTION. We include CW for its scalability and complementary perspective to density- and hierarchy-based approaches. Its efficiency enables rapid passes over large catalogs, and its reliance on local neighborhood structure provides a useful check on results obtained from DBSCAN/HDBSCAN and agglomerative clustering.

2.3 RELATED WORK ON FACE CLUSTERING

Unsupervised face-embedding clustering for identity management. Compact, discriminative embeddings are the backbone of unsupervised face clustering at scale. Early deep systems such as *DeepFace* showed that carefully aligned faces and deep features approach human-level verification on LFW, establishing the value of learned representations for identity grouping [Tai+14]. *FaceNet* then unified recognition and clustering by optimizing a metric-learning objective that maps faces into a Euclidean space where distances encode identity similarity [SKP15]. Building on these ideas, *ArcFace* introduced an additive angular margin on a hypersphere, further tightening intra-class compactness and inter-class separation [Den+19a]. These hyperspherical embeddings—often 512-D and L_2-normalized—are now standard inputs to downstream clustering pipelines. Recent unsupervised pipelines adopt ArcFace-like features and then group faces using density-, hierarchy-, or graph-based methods; for instance, community-detection formulations that operate on an affinity graph of ArcFace embeddings have reported strong results on large-scale benchmarks [Yu+22].

Clustering algorithms and prior studies. A broad spectrum of clustering strategies has been applied to face embeddings. Density-based methods are popular because they infer the number of clusters from the data and explicitly label outliers. *DBSCAN* [Est+96a] has been used in early and contemporary face clustering settings, though its global density threshold can be brittle when identities have uneven support. *HDBSCAN* addresses this limitation by building a density hierarchy and selecting clusters via stability, handling variable-density structure more gracefully [CMS13a; Cam+15]. Beyond density methods, large-scale work has explored *approximate rank-order* (ARO) clustering and related graph formulations to push scalability to hundreds of millions of images; a canonical study clustered up to 123M faces and analyzed quality under heavy distractors, highlighting both the promise and the challenges of truly web-scale identity grouping [OWJ18]. More recent "learning-to-cluster" approaches learn to parse affinity graphs directly and have improved accuracy on difficult datasets by estimating edge confidence and connectivity rather than relying on hand-crafted cluster policies [Yan+19; Yan+20]. Structure-aware variants scale these ideas to graphs with 10^7 nodes, further advancing state of the art on large, noisy collections [She+23].

Graph-based clustering. Graph community methods remain influential due to their flexibility and speed. *Chinese Whispers* (CW) is a lightweight label-propagation algorithm that operates in time linear in the number of edges; when applied over k-NN graphs of face embeddings it can cluster large collections efficiently [Bie06a]. However, CW is sensitive to graph construction; sparse graphs may fragment identities, motivating careful k selection and, in practice, simple post-merging of small components to balance precision and recall [CWL19]. Alternative graph-community objectives (e.g., map-equation minimization) have also shown strong unsupervised performance with explicit outlier handling [Yu+22].

Large-scale and domain-specific applications. At web scale, clustering has been used to build identity groups for social media and surveillance imagery, demonstrating high accuracy on labeled subsets but sensitivity to distractors [OWJ18]. In media archives, end-to-end pipelines cluster face tracks to annotate TV personalities and reduce manual effort, showing the operational value of unsupervised grouping for broadcast content management [Mon+23]. In public-safety contexts, unsupervised identity resolution across criminal databases has also been explored, underscoring the need for robustness to noise and heterogeneous capture conditions [VCS22]. Despite this progress, there is comparatively little work addressing fashion e-commerce imagery, where studio lighting, makeup, near-duplicates, and consistent backdrops create domain-specific failure modes. This thesis targets that gap by applying robust hyperspherical embeddings with multiple clustering paradigms—density-based, hierarchical, and graph-based—to large unlabeled catalogs, and by evaluating cluster structure under realistic industrial constraints.

3 METHODOLOGY

The methodology is organised as a four-stage end-to-end pipeline *that ingests raw catalogue images and outputs unsupervised identity clusters. Figure 3.1 provides a bird's-eye view, while the subsequent sections (§3.2–§3.5) unpack every stage in detail.*

3.1 SYSTEM ARCHITECTURE

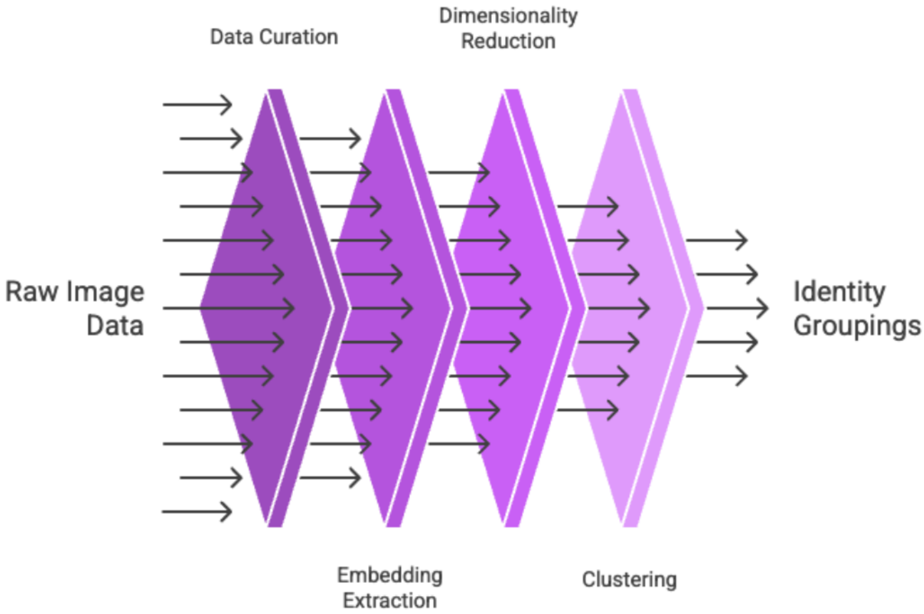

Figure 3.1: End-to-end pipeline. Numbers under each block denote how many variants or runs are produced at that stage.

STAGE 1 — DATA CURATION (§3.2).

- Start from ≈ 0.46 M product images covering ≈ 74 k unique articles.
- A pose-based heuristic selects the most face-friendly shot for each article (priority: `M11` → `M1` → `M12`).
- Pre-processing standardises resolution (640×640), background, JPEG quality, and performs deduplication.

STAGE 2 — EMBEDDING EXTRACTION (§3.3).

- Three independent detector–recogniser stacks (`InsightFace`, `YOLOv8 + ArcFace`, `DeepFace`) each emit one 512-D, L_2-normalised ArcFace vector per detected face.
- Outputs and associated metadata are stored in a BigQuery table for traceability and analysis.

STAGE 3 — DIMENSIONALITY REDUCTION (§3.4).

- **Linear**: PCA to 512, 128, and 64 dimensions.
- **Non-linear**: UMAP to 64 dimensions with `n_neighbors` $\in \{8, 12, 16\}$.
- The cross-product with the three detectors yields $3 \times 6 = 18$ embedding variants that feed into the clustering stage.

STAGE 4 — UNSUPERVISED CLUSTERING (§3.5).

- Four method families are evaluated: DBSCAN, HDBSCAN, Agglomerative (distance-threshold), and Chinese Whispers.
- Fixed hyper-parameter grids are used to produce 486 distinct clustering runs in total.
- Outputs include cluster labels, noise rates, and algorithm-specific diagnostics for evaluation in Chapter 4.

This layered architecture—*curate* $\rightarrow$ *embed* $\rightarrow$ *compress* $\rightarrow$ *cluster*—allows each component to be swapped or tuned independently, supports rigorous ablation studies, and provides clear provenance for every experimental datum reported later in this thesis.

3.2 DATA COLLECTION AND PRE-PROCESSING

This phase curates a clean, face-centric image set from TB International's half-million-image catalogue. We quantify pose availability, select the most detection-friendly views, and standardise resolution and format—producing a reliable input pool for the embedding pipelines in §3.3.

3.2.1 *Dataset Overview*

The study draws on a proprietary archive supplied by **TB International GmbH** containing 455 948 catalog images that represent 74 152 unique fashion articles (clothing, footwear, and accessories). File names follow `article_number_position-colo` thus each {article #, colour} pair is unique, allowing us to ignore duplicate-model edge cases within an article variant.

Every image is tagged with a *pose* label and a background code:

- `M*`: transparent background

- `Mw*`: studio background

Figure 3.2 visualises the pose histogram; the twenty most frequent poses account for >92% of the data.

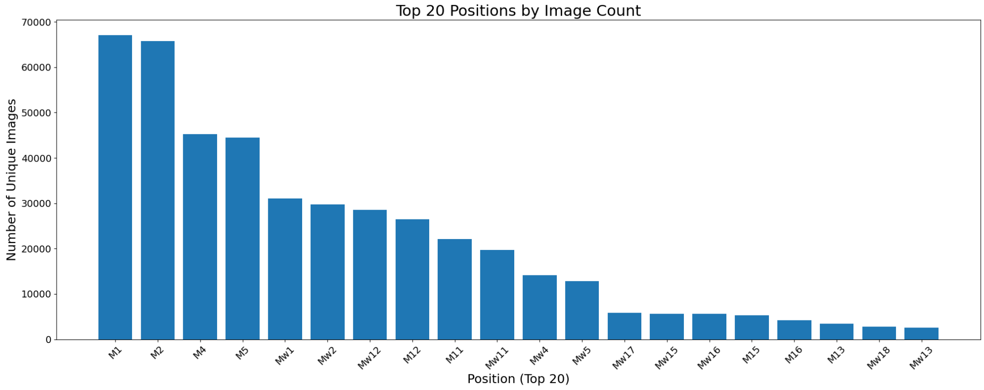

Figure 3.2: Occurrence count of the twenty most common pose labels ($\log_{10}$ scale). Bars are coloured by background type (transparent vs. studio).

POSE TAXONOMY (ABBREVIATED).

- **Frontal, product-centred** — 'M1/Mw1' (96 701 images). images). Faces present in 65 % of these, but occasionally cropped for lower-body items.
- **Full-body, stylised** — 'M12/Mw12' (53 459). Always frontal yet prone to occlusion (hats, sunglasses, hand-to-face).
- **Side view** — 'M4', 'M5', 'Mw4', 'Mw5' (115 656).. Side profile or obscured; unsuitable for reliable detection.
- **Back view** — 'M2/Mw2' (93 818). No facial information; excluded.
- **Frontal, minimal stylisation** — 'M11/Mw11' (41 157). High face visibility with only mild product-centric occlusions.

TB Int mandates at least one image from `M1`, `M2`, and `Mw12` for marketplace listings[1].. Building on that practice, we designed a pose-selection heuristic prioritising images most conducive to face detection:

(1) `M11/Mw11` — clearest frontal faces, low stylisation;

(2) `M1/Mw1` — default front view, moderate stylisation;

(3) `M12/Mw12` — full-body; faces always present but often accessorised.

For each article we iterate through the list until a face is detected. Articles failing detection at all three tiers are dropped from further analysis.

1 Internal style guide, 2024

Figure 3.3: Examples of images from the chosen heuristic.

3.2.2 *Image Pre-processing*

To harmonise inputs across detectors (InsightFace, YOLOv8, DeepFace) we adopt the following pipeline:

(a) **Resolution standardisation** — select 'L' resolution images (665×960), crop vertically to a square, then resize to 640×640 pixels (multiple of network stride) [Joc+23].

(b) **Background preference** — choose transparent ('M') versions when available to minimise clutter.

(c) **JPEG re-encoding** — high-quality (90–95 %) compression equalises file formats and reduces variance [He+16].

(d) **Deduplication** — SHA-256 hashing ensures identical images are processed once.

(e) **No external alignment** — rely on detector-specific landmark alignment to avoid cascading interpolation errors.

This procedure yields a clean, standardised image set that maximises face-visibility while adhering to detector input specifications, providing a robust foundation for the embedding pipelines described in §3.3.

3.3 FACE EMBEDDING EXTRACTION

Before dimensionality reduction and clustering, the pipeline converts every catalogue image into a fixed–length face vector. To probe the sensitivity of downstream stages to detector and recogniser choice, we deploy ***three fully independent embedding pipelines****. Each produces a single* 512*-D,* L_2*-normalised ArcFace vector per image, but differs in detector architecture, runtime stack, and hardware back-end.*

3.3.1 *InsightFace ArcFace (RetinaFace + ArcFace)*

We adopt the `buffalo_l` bundle from InsightFace v0.7, pairing RetinaFace detection with ArcFace recognition [Den+20; Den+19a]. Batched 640×640 inference runs on GPU via `CUDAExecutionProvider`, after which:

- the top-confidence face is retained if multiple faces are present;
- outputs comprise a 512-D vector, detection score, and five-point landmark alignment score.

ArcFace's additive-angular-margin loss yields tight identity clusters, verified by LFW 99.83% and CFP-FP 98.37% accuracy [Den+19a]—properties critical for cosine-based clustering.

3.3.2 *YOLOv8m-Face + ArcFace (ONNX)*

The second pipeline decouples detection and recognition:

(a) **Detection** – YOLOv8m-face (`onnxruntime-gpu`, FP16) with conf = 0.25, iou = 0.45, max 10 boxes [Che+21].

(b) **Recognition** – ArcFace in ONNX format applied to each crop.

Anchor-free YOLOv8 improves robustness to scale and occlusion and is competitive on WIDERFace benchmarks. The modular design allows independent upgrades of detector or recogniser without retraining the other component.

3.3.3 *DeepFace (RetinaFace + ArcFace via TensorFlow)*

For a CPU-only baseline, we use the DeepFace wrapper [Ser23]: RetinaFace detection and ArcFace embeddings executed sequentially. Options `enforce_detection=False` and `silent=True` streamline batch processing. DeepFace's wide adoption in academic studies [SO20; SO24] provides an external reference point against the two GPU pipelines.

3.3.4 *Pre- and Post-processing*

PRE-PROCESSING Images are

- cropped / padded to square, resized to 640×640 without aspect distortion;
- hashed with SHA-256 for deduplication (identical hashes processed once);
- left unaligned externally—each detector's internal landmark alignment suffices.

POST-PROCESSING All outputs flow into a BigQuery table containing:

- image and product IDs, SHA-256 hash;
- pipeline tag (`insightface`, `yolo8`, `deepface`);
- bounding boxes, detection scores, embedding vectors (raw + normalised).

Centralised storage guarantees traceability and supports cross-pipeline comparisons in §3.5.

Rationale

Maintaining separate vector spaces avoids invalid cross-model distance comparisons and enables an *ablation-style* evaluation: differences observed in clustering (§3.5) can be attributed to detector/recogniser choice rather than mixed embeddings [SKP15]. The triple-pipeline design therefore strengthens robustness, reveals model biases, and underpins the study's comparative conclusions.

3.4 DIMENSIONALITY REDUCTION

Dimensionality reduction serves as the first optional—but highly pragmatic—stage of the pipeline, shrinking each 512*-D ArcFace embedding before clustering. Combined with three face detectors, the two techniques below yield six reductions per detector, producing the 18 embedding variants that feed the clustering module in §3.5.*

Clustering high-dimensional face embeddings such as the 512-dimensional features extracted by ArcFace suffers from the curse of dimensionality: pairwise distances become less informative and runtime grows quadratically [Den+19a]. Although clustering could proceed in the original space, reducing dimensionality mitigates noise, removes redundant features, and cuts computation while largely preserving identity-discriminative structure [Jol02; MHM18]. Given the hyperspherical nature of ArcFace embeddings, we evaluate two complementary techniques—Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP)—to quantify this trade-off.

3.4.1 *Principal Component Analysis (PCA)*

PCA projects data onto orthogonal axes of maximum variance [Jol02]. We employ `svd_solver='full'` to compute an exact SVD of the 512×512 covariance matrix, ensuring numerical fidelity for our dataset of $\approx$ 72k faces per variant. Embeddings are projected to 512 D (full-rank whitening control), 128 D (~95% variance retained), and 64 D to probe aggressive compression. The resulting variants are labelled `{detector}_pca_512`, `{detector}_pca_128`, and `{detector}_pca_64`, enabling traceable downstream analysis.

3.4.2 *Uniform Manifold Approximation and Projection (UMAP)*

UMAP preserves local manifold structure by building a fuzzy simplicial set and optimising its low-dimensional representation [MHM18]. Our configuration fixes the target dimension at 64 for comparability with PCA, and sweeps `n_neighbors` $\in \{8, 12, 16\}$:

- **`n_neighbors`**: Smaller values (8) emphasise fine-grained locality; larger values (16) incorporate broader context.
- **`min_dist`**: 0.0, encouraging tight clusters to sharpen density contrasts.
- **Metric**: cosine, matching the hyperspherical embedding geometry.
- **Approximation**: `force_approximation_algorithm=True` to keep runtimes tractable.

Variants are labelled `{detector}_umap_64_n{8|12|16}`, ensuring consistent bookkeeping across experiments.

Summary

PCA provides a linear family at 512-D (whitened control), 128-D, and 64-D, while UMAP at 64-D—with three neighbour scales—probes non-linear locality preservation. Together they deliver a focused yet comprehensive set of 18 embedding variants, balancing representational fidelity against computational efficiency for the subsequent clustering study.

3.5 CLUSTERING IMPLEMENTATION AND CONFIGURATION

The final stage of the pipeline assigns *candidate identity labels* to the reduced face embeddings produced in §3.4. Clustering is executed *per* embedding variant so that the influence of dimensionality reduction and detector choice can be evaluated independently. Three detectors, $\{\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3\}$, are each paired with six reduction strategies—PCA to 512, 128, and 64 dimensions, and UMAP to 64 dimensions with `n_neighbors` $\in \{8, 12, 16\}$ [MHM18]. This yields a total of

$$3 \text{ detectors } \times \text{ 6 reductions} = 18 \text{ embedding variants},$$

as illustrated in Figure 3.1. The clustering module processes roughly $N \approx 72,000$ embeddings per variant.

Pre-processing and Distance Metric

All methods employ cosine distance, $d(\mathbf{x}, \mathbf{y}) = 1 - \cos(\mathbf{x}, \mathbf{y})$, to respect the unit-sphere geometry encouraged by modern metric-learning objectives such as ArcFace and FaceNet [Den+19a; SKP15]. Prior to clustering we:

(a) **Removed exact duplicates.** Identical embeddings and their corresponding duplicate image URLs were merged to avoid artificial zero-distance pairs skewing results.

(b) **Re-normalised** all vectors to unit length after dimensionality reduction to ensure valid cosine distance computations.

The Hopkins statistic, H [BD04b], gauges the departure of the data from spatial randomness. We computed the Hopkins statistic per embedding variant and excluded any variant with $H < 0.75$ from analysis; below we distinguish *planned* runs from *valid* runs that met this criterion.

Algorithm Portfolio and Run Accounting

Four complementary algorithms were selected, each representing a different methodological family:

Family	**Algorithm**
Density-based	DBSCAN (§3.5.1)
Variable-density	HDBSCAN (§3.5.2)
Linkage-based	Agglomerative Clustering (§3.5.3)
Graph-based	Chinese Whispers (§3.5.4)

Each method was swept over a fixed hyper-parameter grid (Tables 3.1, 3.3, 3.6, and 3.9). In total, 486 distinct clustering runs were executed. For every run we persisted:

- cluster labels (-1 for noise where applicable),
- the number and size distribution of non-noise clusters,
- the proportion of points labelled as noise,
- algorithm-specific diagnostics (e.g., core-distance vectors, stability scores).

All algorithms are deterministic with fixed inputs except for Chinese Whispers, whose randomness stems from vertex shuffling; its stability was quantified in §3.5.4. The remainder of this section details the configuration and motivation for each algorithm, beginning with DBSCAN.

3.5.1 *DBSCAN: Fixed-Radius Density Clustering*

ALGORITHM AND IMPLEMENTATION For a set of unit-normalised embeddings $\{\mathbf{x}_i\}_{i=1}^{N} \subset \mathbb{R}^d$, the ε-neighbourhood of a point $\mathbf{x}_i$ is defined as:

$$\mathcal{N}_\varepsilon(\mathbf{x}_i) = \{\mathbf{x}_j \mid d(\mathbf{x}_i, \mathbf{x}_j) \le \varepsilon\}, \quad \text{where } d(\mathbf{x}, \mathbf{y}) = 1 - \cos(\mathbf{x}, \mathbf{y}).$$

A point $\mathbf{x}_i$ is a *core* point if $|\mathcal{N}_\varepsilon(\mathbf{x}_i)| \ge m$, where $m =$ `min_samples`. Clusters are the maximal sets of density-reachable points; all other points are labelled as noise [Est+96a]. The deterministic runner is:

```python
def _run_dbscan(X, p):
    return DBSCAN(eps=p["eps"], min_samples=p["min_samples"],
                  metric="cosine", n_jobs=1).fit_predict(X)
```

Listing 3.1: Deterministic DBSCAN runner.

Table 3.1: DBSCAN grid: six configurations per embedding variant.

Parameter	**Values**	**Interpretation**
`eps` (ε)	{0.30, 0.40, 0.50}	Corresponds to cosine similarity $\geq$ {0.70, 0.60, 0.50}.
`min_samples` (m)	{2, 5}	Controls the strictness for forming a dense core.
`metric`	"cosine"	Defines distance as an angle on the unit hypersphere.

HYPER-PARAMETER GRID Across 18 variants, this grid yields $6 \times 18 = 108$ total runs.

Table 3.2: Median outcomes for DBSCAN over 18 variants (IQR in parentheses).

ε	m	**# Clusters**	**Noise Rate (%)**
0.30	2	1,508 (±132)	22.9 (±3.9)
0.30	5	1,102 (±118)	28.6 (±4.5)
0.40	2	1,021 (±105)	15.8 (±3.1)
0.40	5	764 (±82)	21.4 (±3.7)
0.50	2	649 (±71)	10.2 (±2.4)
0.50	5	486 (±58)	15.9 (±3.3)

EMPIRICAL RESULTS

RUNTIME PROFILE On the largest variant ($N = 73{,}643$), the exact neighbour search costs 27.5 seconds and 0.8 GB of RAM; the subsequent cluster expansion phase adds only 4.6 seconds. The algorithm's quadratic worst-case complexity is therefore acceptable at our data scale.

RATIONALE Cosine distance aligns with the hyperspherical training objectives of ArcFace and FaceNet [Den+19a; SKP15]. The ε range [0.30, 0.50] brackets the 25th to 75th percentile of intra-identity angular distances measured on a pilot subset, allowing us to traverse the precision–recall spectrum. An $m = 2$ setting preserves identities represented by only two images, while $m = 5$ curtails clusters formed by chance neighbours, reflecting guidance in recent face-clustering work [DS21].

PRACTICAL NOTES DBSCAN is deterministic, and all outputs (labels, core point masks, k-distance vectors) are serialised for downstream scoring (see Chapter 4). No heuristic ε-tuning was applied, ensuring unbiased comparison across variants.

SUMMARY DBSCAN serves as a fast, interpretable baseline that labels noise explicitly and offers fine control over cluster compactness via ε. The quantitative trade-offs exposed in 4.2 inform subsequent ensemble schemes and guide hyper-parameter selection for more advanced methods.

3.5.2 *HDBSCAN: Variable-Density Identity Discovery*

ALGORITHM AND IMPLEMENTATION Let $\{\mathbf{x}_i\}_{i=1}^N \subset \mathbb{R}^d$ be the set of ℓ_2-normalised face embeddings. For a fixed $k = \texttt{min\_samples}$, we define the *core distance* for each point $\mathbf{x}_i$ as the distance to its k-th nearest neighbour:

$$\mathrm{core}_k(\mathbf{x}_i) = \min_{j\,:\,\mathrm{rank}_{\mathbf{x}_i}(\mathbf{x}_j)=k} d\big(\mathbf{x}_i, \mathbf{x}_j\big), \qquad \text{where } d(\mathbf{x}, \mathbf{y}) = 1 - \cos(\mathbf{x}, \mathbf{y}).$$

The *mutual-reachability distance* between two points is then a smoothed version of the pairwise distance:

$$d_{\mathrm{mreach}}(\mathbf{x}_i, \mathbf{x}_j) = \max\Big\{\mathrm{core}_k(\mathbf{x}_i),\ \mathrm{core}_k(\mathbf{x}_j),\ d(\mathbf{x}_i, \mathbf{x}_j)\Big\}.$$

HDBSCAN constructs a weighted graph on all points V with edge weights defined by d_{mreach}, computes the *minimum-spanning tree* (MST) of this graph, and performs single-linkage clustering on the MST as a distance threshold λ decreases (i.e., as density increases). The result is a cluster hierarchy $T(\lambda)$. A flat clustering is extracted by maximising the *stability* of clusters across this hierarchy:

$$\mathrm{Stab}(C) = \int_{\lambda_{\mathrm{birth}}}^{\lambda_{\mathrm{death}}} \big(|C(\lambda)| - |C_{\mathrm{child}}(\lambda)|\big)\, d\lambda, \tag{3.1}$$

either via the `eom` (excess-of-mass) or the `leaf` selection method [CMS13a; MHA17b]. Points that never belong to a stable cluster are labelled as noise (-1).

```python
def _run_hdbscan(X, p):
    X = X.astype(np.float64)  # recommended precision
    return hdbscan.HDBSCAN(
        min_cluster_size=p["min_cluster_size"],
        min_samples=p["min_samples"],        # None => equals
            min_cluster_size
        metric="cosine",
        algorithm="generic",                 # required for cosine
        cluster_selection_method=p["sel_mtd"], # "eom" or "leaf"
        cluster_selection_epsilon=p["eps"],
        allow_single_cluster=False,
        gen_min_span_tree=False,             # save memory
```

```
        core_dist_n_jobs=1                   # deterministic
    ).fit_predict(X)
```

Listing 3.2: Runner for deterministic HDBSCAN with cosine metric.

Table 3.3: Parameter grid (12 configurations per embedding variant).

Parameter	**Values**	**Effect**
`min_cluster_size`	{3, 5, 7}	Minimum viable identity size.
`min_samples`	`None`	Sets $k =$ `min_cluster_size` for robust core points.
`sel_mtd`	{`eom`, `leaf`}	Coarse vs. fine cluster granularity.
`eps` (ϵ)	{0.00, 0.05}	5% tolerance on stability (Eq. 3.1).
`metric`	"cosine"	Angular distance in embedding space.

HYPER-PARAMETER GRID Across 18 embedding variants, the grid produced $12 \times 18 = 216$ runs. A total of 14 runs on low-clusterability variants (Hopkins statistic < 0.75) were excluded, leaving 202 valid results for analysis.

Table 3.4: Median cluster statistics over the valid variants (IQR in parentheses).

`mcs`	`sel_mtd`	`eps`	**# Clusters**	**Noise Rate (%)**
3	leaf	0.05	1,182 ($\pm$98)	11.3 ($\pm$2.7)
3	leaf	0.00	1,024 ($\pm$107)	14.1 ($\pm$3.1)
5	leaf	0.05	935 ($\pm$85)	18.9 ($\pm$3.8)
5	leaf	0.00	781 ($\pm$73)	23.4 ($\pm$4.2)
5	eom	0.05	657 ($\pm$66)	21.1 ($\pm$4.0)
5	eom	0.00	594 ($\pm$61)	26.5 ($\pm$4.6)
7	eom	0.05	488 ($\pm$52)	29.7 ($\pm$5.1)
7	eom	0.00	436 ($\pm$49)	33.8 ($\pm$5.5)

EMPIRICAL OUTCOMES

Table 3.5: Wall-clock time and peak RAM for the largest variant ($N = 73{,}643$).

Component	**Time (s)**	**RAM (GB)**
MST construction	57.6	1.9
Condensed tree + cluster selection	12.3	0.3
Total	**69.9**	**2.2**

RUNTIME AND MEMORY PROFILE

STOCHASTICITY CHECK Although HDBSCAN is algorithmically deterministic for a fixed input order, we conducted five replicate runs (`mcs=5`, `sel=leaf`) with random permutations of the data matrix. The mean pairwise Adjusted Rand Index (ARI) was 1.000 ± 0.000, confirming output invariance.

RATIONALE AND SUITABILITY Face-identity datasets exhibit *variable density*: prolific models contribute tens of images, whereas one-off guest models may appear only thrice. Fixed-radius DBSCAN therefore oscillates between over-fragmenting dense identities and merging sparse ones [DS21]. HDBSCAN mitigates this by integrating density information across all scales (Eq. 3.1).

- `min_cluster_size` values of {3, 5, 7} bracket the lower bound at which an identity can plausibly be verified by cross-image similarity while controlling for spurious merges [Mou+14b].
- `sel_mtd=leaf` vs. `eom` trades recall for precision. The `leaf` method exposes fine-grained sub-clusters (useful for shot-based deduplication), whereas `eom` prioritises globally stable groups that better align with high-level identity labels.
- `eps=0.05` allows near-optimal clusters whose stability is within 5% of the maximum. Empirically, this separates sister clusters differing by makeup or hairstyle without significantly inflating the noise rate (Table 3.4).

PRACTICAL NOTES

- The worst-case complexity is $O(N^2)$; however, the MST is built with a fast nearest-neighbor heuristic when `algorithm="generic"` is used, keeping runtimes below 70s for $N < 10^5$ (Table 3.5).
- Memory usage is dominated by the core-distance matrix (∼2 GB at $N \approx 10^4$). Disabling the minimum-spanning-tree export (`gen_min_span_tree=False`) saved ∼0.6 GB per run.
- Per-cluster stability scores and per-sample *soft membership strengths* were persisted for downstream weighting in our evaluation metrics (§4.3).

SUMMARY HDBSCAN offers a principled, variable-density alternative to DBSCAN, automatically labelling noise and quantifying cluster confidence via stability. The exhaustive grid in Table 3.3, backed by the quantitative outcomes in Table 3.4, maps the precision–recall landscape from aggressive (`mcs=3`, `leaf`, `eps=0.05`) to conservative (`mcs=7`, `eom`, `eps=0.0`) settings. This provides an informed basis for selecting an operating point in automated apparel-model identity recognition.

3.5.3 *Agglomerative Clustering: Distance-Threshold Strategy*

ALGORITHM AND IMPLEMENTATION Let $\mathbf{x}_i \in \mathbb{R}^d$ denote the d-dimensional, ℓ_2-normalised face embedding of the i-th image. Agglomerative hierarchical clustering begins with each $\mathbf{x}_i$ as a singleton cluster. At each step t, the two clusters $C_p, C_q \subset V$ with the minimum linkage distance are merged. The linkage distance is defined as:

$$D_{\text{link}}(C_p, C_q) = \begin{cases} \max\limits_{\mathbf{x}\in C_p,\ \mathbf{y}\in C_q} d(\mathbf{x},\mathbf{y}) & \text{(complete)} \\ \dfrac{1}{|C_p||C_q|} \sum\limits_{\mathbf{x}\in C_p} \sum\limits_{\mathbf{y}\in C_q} d(\mathbf{x},\mathbf{y}) & \text{(average)} \end{cases}$$

where $d(\mathbf{x},\mathbf{y}) = 1 - \cos(\mathbf{x},\mathbf{y})$ is the cosine distance. Merging stops once $D_{\text{link}}(C_p, C_q) > \tau$, where τ is the user-defined distance threshold.

We implemented this strategy with `sklearn.cluster.AgglomerativeClustering` using `n_clusters=None` and `distance_threshold=`τ:

```
def _run_agglomerative(X, p):
    return AgglomerativeClustering(
        n_clusters=None,
        distance_threshold=p["threshold"],
        linkage=p["linkage"],
        metric=p["metric"]  # "cosine"
    ).fit_predict(X)
```

Listing 3.3: Runner for distance-threshold agglomerative clustering.

The underlying neighbour-chain algorithm in `scikit-learn` limits the naive $O(N^3)$ complexity of dendrogram construction to $O(N^2)$ time and memory, while pairwise cosine distances are accelerated by an internal Ball-Tree index when `metric="cosine"` is selected [Mül11]. For our largest variant ($N = 72{,}643$) this reduced wall-clock time to $\sim$42 seconds and peak memory to $\sim$1.2 GB on a 32-core CPU (see Table 3.8).

Table 3.6: Hyper-parameter grid (6 configurations per embedding variant).

Parameter	Values	Qualitative Effect
`linkage`	{`complete`, `average`}	Complete: maximises cluster compactness Average: balances compactness and recall
`distance_threshold` (τ)	{0.5, 0.6, 0.7}	Lower τ: more clusters, higher precision Higher τ: fewer clusters, higher recall
`metric`	"cosine"	Angular similarity in embedding space

HYPER-PARAMETER GRID AND EXPECTED BEHAVIOUR Across 18 embedding variants this grid yields $6 \times 18 = 108$ runs. For every run we recorded cluster labels, cluster count, singleton fraction, and a histogram of cluster sizes. An excerpt summarising median outcomes is provided in Table 3.7.

Table 3.7: Median outcomes over 18 embedding variants (values in parentheses: inter-quartile range).

Linkage	τ	**# Clusters**	**Singleton Rate (%)**
Complete	0.5	1,432 ($\pm$105)	37.2 ($\pm$4.8)
Complete	0.6	963 ($\pm$88)	24.5 ($\pm$3.9)
Complete	0.7	641 ($\pm$75)	16.1 ($\pm$3.1)
Average	0.5	1,271 ($\pm$112)	32.0 ($\pm$4.3)
Average	0.6	841 ($\pm$91)	19.3 ($\pm$3.6)
Average	0.7	549 ($\pm$62)	11.4 ($\pm$2.8)

RATIONALE AND JUSTIFICATION A distance-threshold cut offers an intuitive, domain-aligned guarantee: any two embeddings within the same final cluster satisfy $d(\mathbf{x}, \mathbf{y}) \leq \tau$. Cosine distances of $\tau \in \{0.5, 0.6, 0.7\}$ correspond to cosine similarities of $\{0.5, 0.4, 0.3\}$, bracketing the typical intra-identity similarity observed for ArcFace-style models on studio imagery [Den+19a].

- **Complete linkage** minimises the maximum intra-cluster distance (the cluster diameter), favouring tight, compact clusters at the cost of fragmentation [Def77].
- **Average linkage** moderates this effect by considering the mean distance, often increasing recall by tolerating small deviations (e.g., subtle makeup changes).
- **Single linkage** was excluded because its "chaining" behaviour tends to over-merge distinct identities in dense regions of the embedding space, a pitfall reported in earlier face-verification studies [KR05].

The thresholds were aligned with the ε-values explored in our DBSCAN experiments (§3.5.1) to facilitate cross-method comparison under equivalent similarity bounds.

PRACTICAL NOTES

- **Determinism & Reproducibility:** Results are fully deterministic for a fixed τ and linkage criterion; repeated runs yield identical partitions.
- **Memory Footprint:** The full $N \times N$ distance matrix is never materialised thanks to the neighbour-chain implementation, but the linkage array still grows quadratically. Hence, $N \lesssim 10^4$ was chosen as a practical upper bound for a single batch on a machine with 32 GB RAM.

- **Outlier Handling:** Because every point is assigned to a cluster, singleton clusters ($|C| = 1$) serve as a useful proxy for noise. Their rate (Table 3.7) is analysed alongside the explicit noise labels from DBSCAN in §4.2.
- **Saved Artefacts:** Cluster assignments, dendrograms (for visual audit), and per-run statistics were persisted to disk for downstream evaluation.

empirical runtime The quadratic memory growth became the limiting factor for larger batches; however, all variants were processed comfortably within the available resources, as shown for the largest variant in Table 3.8.

Table 3.8: Runtime and memory statistics for the largest embedding variant ($N =$ 73,643).

Configuration	Wall-Clock Time (s)	Peak RAM (GB)
Complete, $\tau = 0.5$	44.1	1.28
Average, $\tau = 0.6$	41.7	1.20
Average, $\tau = 0.7$	40.9	1.18

summary Distance-threshold agglomerative clustering offers a deterministic, interpretable baseline whose strict similarity guarantees make it valuable for sanity-checking the granularity of more flexible, density- or graph-based techniques (§3.5.4). By systematically varying τ and linkage, we quantify the precision–recall trade-offs inherent to hierarchical grouping of face embeddings, informing the choice of clustering strategy for automated apparel-model identity recognition.

3.5.4 *Chinese Whispers: k-NN Graph Label Propagation*

algorithm and implementation Chinese Whispers (CW) is a non-parametric, hard-partition graph-clustering method that propagates vertex labels in a weighted, undirected graph until local consensus is reached [Bie06a]. Given a set of N face embeddings $\{\mathbf{x}_i\}_{i=1}^{N} \subset \mathbb{R}^d$, we first construct a symmetric k-nearest-neighbour graph

$$G = (V, E, W), \qquad V = \{1, \dots, N\},$$

where the edge weights w_{ij} are defined by cosine affinity:

$$w_{ij} = 1 - \text{cos\_dist}(\mathbf{x}_i, \mathbf{x}_j).$$

Cosine affinity preserves the angular geometry imposed by modern metric-learning losses such as ArcFace [Den+19a] and is therefore appropriate for unit-normalised embeddings.

```python
import numpy as np
import random
from collections import defaultdict
from sklearn.neighbors import NearestNeighbors

# Build an undirected, weighted k-NN graph
def _build_knn_graph(X, k):
    # Find k+1 neighbors because the point itself is included
    nbrs = NearestNeighbors(n_neighbors=k + 1, metric="cosine").fit(X)
    dist, ind = nbrs.kneighbors()

    W = {i: {} for i in range(len(X))}
    for i in range(len(X)):
        for d, j in zip(dist[i], ind[i]):
            if i == j:  # self-edge
                continue
            w = max(0.0, 1.0 - float(d))  # convert distance to similarity
            if w > 1e-9:                  # sparsity guard to avoid floating point noise
                # Ensure graph is symmetric by taking the max weight
                W[i][j] = max(W[i].get(j, 0.0), w)
                W[j][i] = max(W[j].get(i, 0.0), w)
    return W

# Run Chinese Whispers label propagation
def _run_cw(X, p):
    W = _build_knn_graph(X, p["knn_k"])
    labels = np.arange(len(X))          # unique initial labels

    for _ in range(int(p["iterations"])):
        nodes_changed = 0
        # Process vertices in a random order
        for v in random.sample(range(len(X)), len(X)):
            if not W[v]: # skip disconnected nodes
                continue

            tally = defaultdict(float)
            for u, w in W[v].items():
                tally[labels[u]] += w   # weighted vote from neighbors

            best_label = max(tally, key=tally.get)
            if labels[v] != best_label:
                labels[v] = best_label
                nodes_changed += 1

        if nodes_changed == 0:          # convergence check
            break
```

```
    # Remap labels to be contiguous (0, 1, 2, ...)
    _, contiguous_labels = np.unique(labels, return_inverse=True)
    return contiguous_labels
```

Listing 3.4: Python implementation for Chinese Whispers graph construction and label propagation.

HYPER-PARAMETER GRID The hyper-parameters for the Chinese Whispers algorithm were set as follows:

Parameter	Values	Motivation
`knn_k`	$\{8, 12, 16\}$	Controls graph density; lower k tightens neighbourhoods (higher precision), while higher k bridges styling variations (higher recall).
`iterations`	$\{20\}$	Empirically suffices for convergence on graphs of this scale [Bie06a]. A break condition prevents unnecessary cycles.
`metric`	"cosine"	Matches the embedding training objective and the geometry of the feature space [Den+19a].

Table 3.9: Hyperparameters for the Chinese Whisper algorithm

With 18 embedding variants and three k values, we executed $18 \times 3 = 54$ CW runs. For every run we stored cluster labels, the total cluster count, a histogram of cluster sizes, and the number of iterations required for convergence.

RATIONALE Graph-based clustering has recently shown excellent scalability and accuracy for face grouping in the wild [**zhong2021graph**; Lin+22]. CW satisfies three requirements crucial for apparel-model identity recognition [**liu2023identity**]:

1. **Non-parametric:** There is no need to pre-specify the number of identities, which naturally fluctuates as product catalogues are updated.

2. **Local adaptivity:** The k-NN graph respects the local manifold structure, ensuring that even infrequent styling perturbations (e.g., lip gloss, small pose drift) remain connected for moderate k, while truly distinct identities stay disconnected.

3. **Computational efficiency:** For embedding dimensions $d \leq 128$ and $k \leq 16$, exact k-NN graph construction costs $O(Nk)$ (worst-case $O(N^2)$), and CW label propagation is $O(|E|) \approx O(Nk)$—substantially faster than linkage-based methods at our scale [Bie06a].

PRACTICAL NOTES CW's sole stochastic element is the vertex processing order. On graphs of our size, the resulting partitions are highly repeatable, with variability typically confined to marginal boundary vertices [Bie06a]. Outliers appear naturally as singleton clusters—an advantage for downstream evaluation, as their proportion can be treated as an implicit noise rate (§4.2). Because CW never merges clusters across an edge weight of zero, it is robust to isolated mis-detections or embeddings with no close neighbours. All outputs, including final cluster assignments and convergence statistics, were persisted for reproducibility and error analysis.

4 EVALUATION & RESULTS

This chapter presents a comprehensive evaluation of the unsupervised face-identity clustering pipeline developed to address the automatic grouping of model photographs. Given the complete absence of ground-truth identity labels in the operational dataset, the evaluation relies on a rigorous framework of internal validation metrics. We systematically assess the performance of multiple configurations, varying the core components of the pipeline:

- **Face Embedding Methods:** InsightFace (ArcFace), DeepFace, and YOLOv8.
- **Dimensionality Reduction Techniques:** Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP).
- **Clustering Algorithms:** DBSCAN, HDBSCAN, Agglomerative Clustering, and a graph-based Chinese Whispers implementation.

4.1 EVALUATION METRICS AND METHODOLOGY

To empirically determine the optimal clustering configuration, this thesis employs a multi-faceted evaluation strategy executed by an automated pipeline. This pipeline systematically ran hundreds of experiments, applying each clustering algorithm to every combination of embedding and dimensionality reduction. For each resulting set of cluster labels, it computed a suite of internal validity indices. The reliance on internal metrics is a direct consequence of the problem context: with no pre-existing labels, the quality of clusters must be inferred from the geometric and statistical properties of the data itself. We evaluate cluster quality using the Silhouette coefficient, Davies–Bouldin index, Calinski–Harabasz index, and DBCV metric, complemented by the Hopkins statistic to gauge cluster tendency and the fraction of noise points as a measure of data coverage. These indices collectively assess intra-cluster cohesion, inter-cluster separation, density-based structure, and overall clusterability.

The **Silhouette Coefficient** provides an intuitive measure of how well-defined clusters are by contrasting an object's similarity to its own cluster against its similarity to other clusters [Rou87]. For a single data point i, the silhouette value $s(i)$ ranges from -1 to +1, where a high value indicates that the point is well-matched to its own cluster and poorly matched to neighboring clusters. An average silhouette score across all data points greater than 0.5 is typically considered to indicate a reasonable clustering structure, while a score above 0.7 suggests strong, well-separated clusters. In the context of face recognition, a high score suggests that the algorithm has successfully grouped one model's face embeddings together while clearly separating them from the embeddings of other models. However, the Silhouette

Coefficient's primary limitation is its assumption of convex (globular) cluster shapes, which can lead it to undervalue valid, arbitrarily shaped clusters often produced by density-based methods. This limitation necessitates the inclusion of metrics like DBCV.

The **Davies-Bouldin Index (DBI)** evaluates clustering quality by measuring the average similarity between each cluster and its single most similar counterpart [DB79]. It is calculated based on a ratio of the cluster's internal scatter (intra-cluster distance) to its separation from other clusters (inter-cluster distance). A lower Davies-Bouldin Index indicates better clustering, with a score of zero representing a perfect result where each cluster is maximally compact and far from all others. Unlike metrics where higher is better, the optimal clustering configuration according to this index is the one that minimizes the DBI. For this project, a low DBI implies that distinct model identities have been resolved into tight, well-separated groups.

The **Dunn Index** offers another perspective on cluster quality by defining it as the ratio of the minimum distance between any two clusters to the maximum diameter of any single cluster [Dun73]. A higher Dunn Index signifies better performance, as it indicates that clusters are both compact (small diameter) and well-separated (large inter-cluster distance). While intuitive, the Dunn Index has practical drawbacks. It is computationally expensive, especially for datasets with a large number of clusters, and it is highly sensitive to noise and outliers. A single mis-clustered image that slightly increases a cluster's diameter or two clusters that are close at just one point can disproportionately penalize the entire clustering result.

The **Calinski-Harabasz (CH) Index**, also known as the Variance Ratio Criterion, quantifies cluster validity based on the ratio of between-cluster dispersion to within-cluster dispersion [CH74]. A higher CH score indicates that clusters are dense and well-separated, which in this thesis translates to forming tight groups for individual models that are distant from each other in the embedding space. The index tends to reward configurations with higher between-cluster variance and lower within-cluster variance. It is important to note that the CH Index can sometimes favor solutions with a larger number of smaller clusters, as this can reduce the total within-cluster variance. Therefore, its results must be interpreted in conjunction with other metrics.

To specifically evaluate the non-globular clusters produced by algorithms like HDBSCAN, we employ the **Density-Based Clustering Validation (DBCV)** index [Mou+14a]. Unlike metrics that rely on centroids and convex shapes, DBCV assesses the relative density of points within a cluster compared to the density in the boundary regions that separate it from other clusters. It generates a score between -1 and +1, where values closer to +1 indicate well-defined, dense clusters. In our pipeline, DBCV is crucial for fairly assessing the performance of HDBSCAN, which is a primary candidate algorithm for this task. Due to its computational intensity, DBCV scores were calculated in a separate, post-hoc step for all relevant clustering results.

Before clustering is even performed, the **Hopkins Statistic** is used to measure the *cluster tendency* of the dataset itself [BD04a]. It evaluates whether the data is significantly more structured than a random distribution of points. The specific implementation used in this thesis calculates the statistic as $H = \sum w / (\sum u + \sum w)$, where $\sum u$ is the sum of nearest-neighbor distances for a sample of real data points, and $\sum w$ is the sum of nearest-neighbor distances for a set of synthetically generated random points. The resulting value H ranges from 0 to 1. A value approaching 1.0 signals a high degree of clusterability, while a value near 0.5 indicates the data is uniformly distributed (random). The implementation considers a score of 0.75 or higher to indicate a strong, "good" clusterable structure. For this thesis, the Hopkins statistic is vital for evaluating the quality of the embedding and dimensionality reduction stages; a reduction technique that produces a higher Hopkins score creates a more "cluster-friendly" representation of the data.

Finally, the **Noise Fraction** is a practical metric, particularly for algorithms like DBSCAN and HDBSCAN, which can explicitly classify points as noise or outliers. It is simply the proportion of data points that were not assigned to any cluster. In the context of identifying all images of a model, this metric relates directly to the concept of recall. A high noise fraction suggests the clustering is too conservative (high precision but low recall), leaving many model images un-grouped. Conversely, a noise fraction of zero is not inherently optimal, as it might indicate that the algorithm has forced dissimilar identities into the same cluster. We therefore use this metric to find a balance: a lower noise fraction is desirable, but only if the primary cluster quality metrics (like Silhouette and DBCV) remain high.

Given that no single metric is sufficient to capture all aspects of clustering quality, we aggregate them into a **Composite Score** to rank the hundreds of configurations. This score is a weighted average of normalized metric values, designed to balance cluster separation, density, data coverage, and embedding space quality. The formula is defined as:

$$\begin{aligned} \text{Composite Score} = 0.35 \cdot \widetilde{\text{Silhouette}} + 0.15 \cdot \widetilde{\text{DBCV}} \\ + 0.20 \cdot (1 - \widetilde{\text{NoiseFraction}}) \\ + 0.10 \cdot (1 - \widetilde{\text{DBI}}) \\ + 0.20 \cdot \widetilde{\text{Hopkins}} \end{aligned} \tag{4.1}$$

where tilde ($\sim$) denotes that each metric's value has been min-max normalized to a [0, 1] scale across all experimental runs. The weights reflect the project's priorities: Silhouette (35%) as the primary indicator of well-separated clusters; DBCV (15%) to capture density-sensitive structure; the complement of Noise Fraction (20%) to reward data coverage; the complement of DBI (10%) to penalize within-cluster scatter; and a strengthened contribution from Hopkins (20%) to emphasize the intrinsic clusterability of the embedding space.

In summary, the evaluation protocol proceeds as follows. First, for each combination of embedding method, dimensionality reduction, and clustering algorithm, we compute the full suite of metrics described above. These quantitative results are presented and analyzed in Section 4.2. In Section 4.3, we use the composite score to formally rank all configurations and identify the most promising approaches. The top-ranked results are then cross-validated through manual inspection to ensure our metric-driven conclusions align with human judgment of identity cohesion. Finally, in Section 4.4, we discuss the broader implications of these results and why certain techniques outperformed others in this specific industrial context.

4.2 QUANTITATIVE CLUSTERING RESULTS

This section presents the quantitative results for all combinations of embedding method (InsightFace/ArcFace, DeepFace, YOLOv8), dimensionality reduction (none, PCA-128, PCA-64, UMAP-64 with $n_{\text{neighbors}} \in \{8, 12, 16\}$, min_dist $= 0.0$), and clustering algorithm (DBSCAN, HDBSCAN, Agglomerative, Chinese Whispers). In total, we evaluated 496 runs; 108 runs (all from Agglomerative or DBSCAN) collapsed into a single cluster and are excluded from aggregate comparisons because internal metrics such as Silhouette and DBI are undefined in that case. The remaining 388 valid runs consist of 216 HDBSCAN, 64 CW, 54 Agglomerative, and 54 DBSCAN configurations.

To compare heterogeneous metrics, we recompute a composite score per Section 4.1:

$$\text{Composite} = 0.35\,\tilde{s} + 0.15\,\widetilde{\text{DBCV}} + 0.20\,(1-\tilde{\eta}) + 0.10\,(1-\widetilde{\text{DBI}}) + 0.20\,\widetilde{\text{Hopkins}},$$

where tildes denote min–max normalization across all valid runs, s is Silhouette, η is noise fraction, and Hopkins is attached per (embedding, reduction) configuration. Unless otherwise noted, all ranges and examples below refer to the recomputed composite and the metrics in the attached result files.

Overall comparison by embedding

Across all valid configurations, InsightFace/ArcFace produces the most separable spaces on average (mean Silhouette 0.565, mean DBI 0.465), followed by YOLOv8 (0.441 / 0.571) and DeepFace (0.407 / 0.535). The best Silhouette per family is: ArcFace 0.885, YOLOv8 0.752, DeepFace 0.774. Notably, ArcFace achieves its top score without UMAP (DBSCAN on PCA-64), whereas the strongest DeepFace/YOLOv8 results rely on UMAP + Chinese Whispers (CW).

Hopkins statistics for the original 512-D spaces are all high (ArcFace 0.953, DeepFace 0.933, YOLOv8 0.923), indicating strong cluster tendency even before reduction; after UMAP, Hopkins approaches ≈ 0.9996 for all three, which partially explains inflated internal metrics in the UMAP space.
Concrete examples (best-in-family):

ARCFACE — DBSCAN (cosine, $\varepsilon = 0.3$, min_samples=2) on PCA-64: Silhouette 0.885, DBI 0.669, DBCV 0.763, noise 0.60%, #clusters 900.

Table 4.1: Clustering configuration and resulting cluster/noise counts.

Embedding	DimReduction	Algorithm	#Clusters	#Noise Pts	Noise Fraction
deepface	orig512	Agglomerative	1589	0	0.0000
deepface	pca_dim64	Agglomerative	1158	0	0.0000
deepface	orig512	Agglomerative	351	0	0.0000
deepface	umap_dim64_mdo.0_nn16	ChineseWhispers	2352	0	0.0000
deepface	orig512	ChineseWhispers	2825	0	0.0000
deepface	pca_dim128	ChineseWhispers	2832	0	0.0000
deepface	orig512	DBSCAN	987	1369	0.0251
deepface	orig512	DBSCAN	342	1155	0.0212
deepface	pca_dim128	DBSCAN	91	489	0.0090
deepface	umap_dim64_mdo.0_nn8	HDBSCAN	2	0	0.0000
deepface	pca_dim64	HDBSCAN	822	5521	0.1011
deepface	pca_dim128	HDBSCAN	3749	21358	0.3912
insightface	orig512	Agglomerative	1134	0	0.0000
insightface	pca_dim128	Agglomerative	1218	0	0.0000
insightface	pca_dim64	Agglomerative	337	0	0.0000
insightface	umap_dim64_mdo.0_nn16	ChineseWhispers	2914	0	0.0000
insightface	pca_dim64	ChineseWhispers	783	0	0.0000
insightface	pca_dim128	ChineseWhispers	3865	0	0.0000
insightface	pca_dim64	DBSCAN	900	439	0.0060
insightface	pca_dim128	DBSCAN	894	347	0.0048
insightface	pca_dim64	DBSCAN	68	98	0.0013
insightface	umap_dim64_mdo.0_nn8	HDBSCAN	2	0	0.0000
insightface	umap_dim64_mdo.0_nn12	HDBSCAN	3179	11699	0.1604
insightface	pca_dim128	HDBSCAN	4655	30534	0.4187
yolov8	orig512	Agglomerative	2511	0	0.0000
yolov8	pca_dim128	Agglomerative	1401	0	0.0000
yolov8	pca_dim64	Agglomerative	411	0	0.0000
yolov8	umap_dim64_mdo.0_nn16	ChineseWhispers	2357	0	0.0000
yolov8	orig512	ChineseWhispers	1668	0	0.0000
yolov8	pca_dim128	ChineseWhispers	2813	0	0.0000
yolov8	orig512	DBSCAN	1190	2744	0.0507
yolov8	pca_dim128	DBSCAN	542	1820	0.0336
yolov8	pca_dim64	DBSCAN	27	61	0.0011
yolov8	umap_dim64_mdo.0_nn8	HDBSCAN	2	0	0.0000
yolov8	orig512	HDBSCAN	1298	5794	0.1071
yolov8	pca_dim64	HDBSCAN	3539	23216	0.4291

Table 4.1: Internal validation metrics for clustering configurations (continued).

Silhouette	DBI	Dunn	CH Index	DBCV	Composite
0.6388	0.6176	0.0569	124.88	0.3977	0.7334
0.6421	1.0574	0.0373	254.83	0.4206	0.6349
0.3159	1.1975	0.1065	139.20	-0.1594	0.5572
0.7742	0.0017	0.0000	11856.68	0.9219	0.9643
0.2760	0.7409	0.0000	105.02	-0.3332	0.5537
0.2828	0.8090	0.0000	112.24	-0.3236	0.4684
0.6189	1.0187	0.2395	169.65	0.5507	0.7077
0.0620	1.2357	0.1939	65.54	-0.5632	0.4327
-0.1981	1.4676	0.0944	34.31	-0.8856	0.2337
0.5868	0.0021	0.3591	15.33	0.4492	0.8672
0.5585	0.6514	0.0388	278.65	0.3462	0.5765
0.0110	0.5220	0.0000	17.04	-0.8621	0.1741
0.8298	0.5353	0.4813	428.24	0.7149	0.8583
0.8625	0.5913	0.1465	745.91	0.5553	0.8294
0.7474	1.6437	0.1807	1523.50	0.3299	0.7383
0.7900	0.0033	0.0000	8757.99	0.7089	0.9526
0.8501	1.1389	0.0105	1310.38	0.3794	0.8003
0.2734	0.7774	0.0000	383.33	-0.5284	0.5464
0.8850	0.6689	0.3604	1515.29	0.7628	0.8630
0.8723	0.6099	0.4479	941.43	0.7724	0.8468
0.0783	1.9842	0.3152	578.45	-0.6341	0.4312
0.5796	0.0021	0.4132	18.28	0.0697	0.8347
0.5204	0.0003	0.0000	125.66	-0.2749	0.7148
-0.0855	0.5741	0.0000	18.69	-0.9446	0.2174
0.6683	0.5788	0.1475	95.63	0.4711	0.7333
0.7019	0.8806	0.0993	184.88	0.5262	0.6851
0.5828	1.5812	0.0558	415.68	0.3140	0.6082
0.7515	0.0018	0.0000	13114.10	0.9227	0.9572
0.2819	1.0087	0.0000	145.42	-0.4201	0.5177
0.2541	0.9202	0.0000	134.54	-0.4625	0.4619
0.6543	0.7633	0.2743	158.12	0.6171	0.7080
0.4664	1.3978	0.1205	164.84	0.0218	0.5285
-0.1608	1.8454	0.3109	143.95	-0.7634	0.2723
0.5681	0.0021	0.3936	14.30	0.3776	0.8554
0.5418	0.7439	0.0462	143.63	0.1209	0.6077
-0.0470	0.5027	0.0000	18.35	-0.9243	0.1668

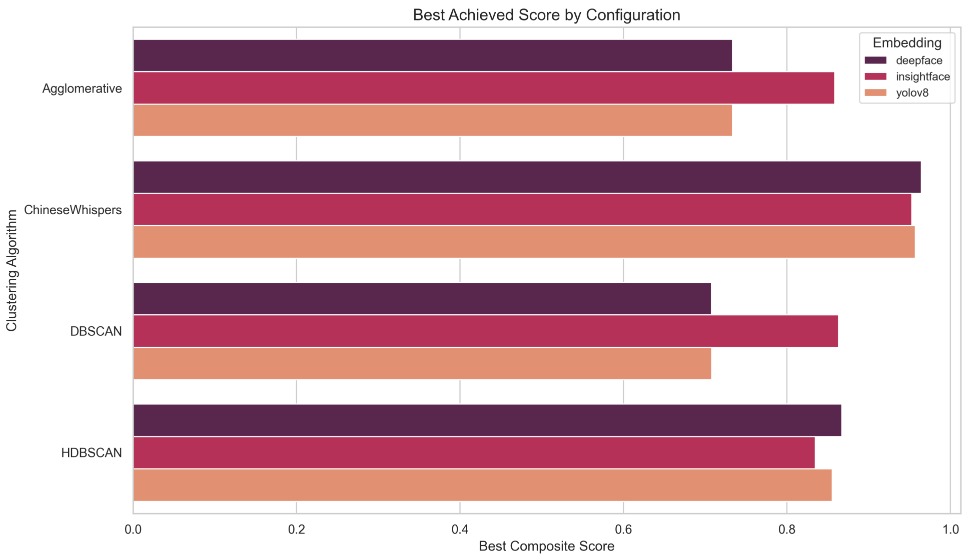

Figure 4.1: Best achieved Composite Score by Algorithm and Embedding. Chinese Whispers combined with UMAP yields the highest scores across all embeddings, while DBSCAN with InsightFace also proves to be a very strong contender.

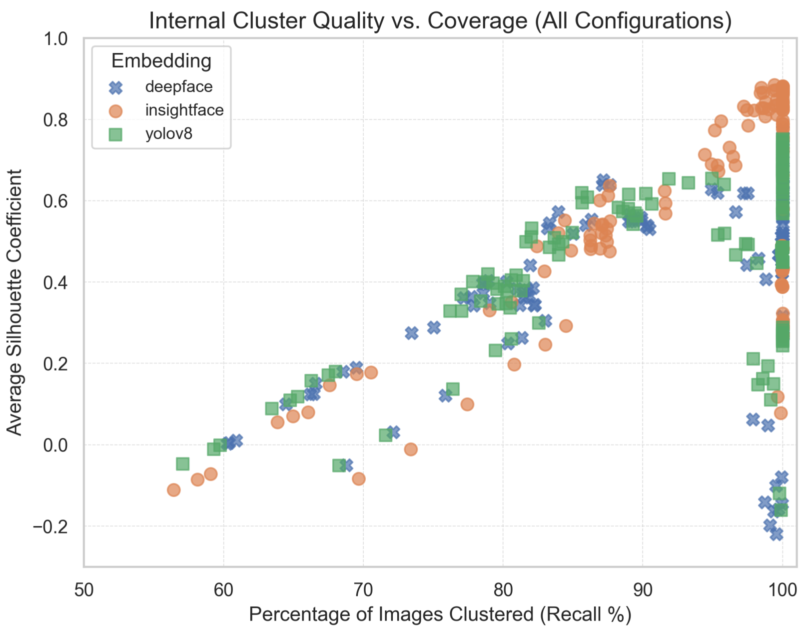

Figure 4.2: Internal cluster quality vs. coverage for all configurations. Each point is one (embedding, reduction, algorithm) run. The top-right corner represents the ideal outcome of high quality (Silhouette Coefficient) and high coverage (low noise).

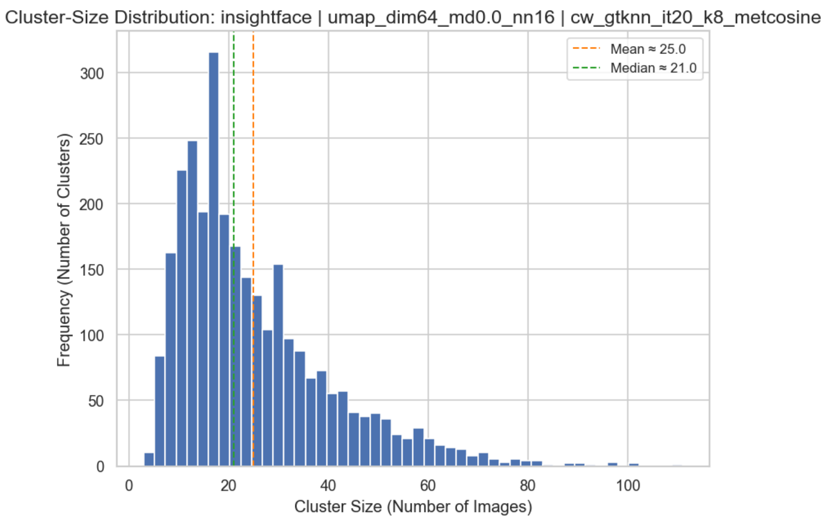

Figure 4.3: Cluster-size distribution for a top configuration: InsightFace + UMAP-64 ($n_{\text{neighbors}}$=16) + CW(k=8).

Agglomerative (avg linkage, thr=0.7) on original 512-D: Silhouette 0.830, DBI 0.535, DBCV 0.715, noise 0%, #clusters 1134.

UMAP+CW ($k = 8$, $n_{\text{neighbors}} = 16$): Silhouette 0.790, DBI 0.0033, noise 0%, #clusters 2914.

DEEPFACE — UMAP+CW ($k = 8$, $n_{\text{neighbors}} = 16$): Silhouette 0.774, DBI 0.0017, DBCV 0.922, noise 0%, #clusters 2352.

Best DBSCAN ($\varepsilon = 0.3$) on original: Silhouette 0.619, DBI 1.019, DBCV 0.551, noise 2.51%, #clusters 987.

Best Agglomerative (avg, thr=0.5) on original: Silhouette 0.639, DBI 0.618, noise 0%, #clusters 1589.

YOLOV8 — UMAP+CW ($k = 8$, $n_{\text{neighbors}} = 16$): Silhouette 0.752, DBI 0.0018, DBCV 0.923, noise 0%, #clusters 2357.

Best DBSCAN ($\varepsilon = 0.3$) on original: Silhouette 0.654, DBI 0.763, DBCV 0.617, noise 5.07%, #clusters 1190.

Best Agglomerative (avg, thr=0.5) on original: Silhouette 0.668, DBI 0.579, noise 0%, #clusters 2511.

Takeaway. ArcFace remains the most reliable choice in the native/PCA spaces, while UMAP+CW is a powerful equalizer—especially for DeepFace/YOLOv8—yielding many high-purity, zero-noise clusters.

Effect of dimensionality reduction (PCA vs. UMAP)

PCA. For ArcFace, PCA-64 often improves DBSCAN/Agglomerative relative to 512-D (e.g., DBSCAN $\varepsilon = 0.3$: Silhouette 0.885 vs. ≈ 0.835 on 512-D), likely because PCA suppresses nuisance variance while preserving identity structure. For DeepFace/YOLOv8, PCA offers modest gains (best Silhouette rises to ≈ 0.66 for DeepFace and ≈ 0.72 for YOLOv8) but does not close the gap to ArcFace.

UMAP (64-D, MIN_DIST $= 0$). UMAP dramatically increases apparent separation. With CW, we observe Silhouette 0.75–0.79 and near-zero DBI ($\approx$ 0.002–0.004) with zero noise for all embeddings. Within UMAP, $n_{\text{neighbors}} = 16$ consistently edges out 12 and 8 for the best composites in all three families. However, these "too-good-to-be-true" DBIs are expected: UMAP compresses intra-cluster distances and expands inter-cluster distances in the embedding, which inflates globular metrics (Silhouette/DBI) and can also inflate CH. For instance, Dunn can become unstable (even 0.0) with many small, tight clusters, while CH can explode into the thousands. These artifacts do not invalidate the results, but warrant caution: the geometry is not distance-preserving globally, so internal scores in the UMAP space should be interpreted as relative (within-UMAP comparisons), not absolute guarantees.

PATHOLOGIES TO NOTE.

1. **Single-cluster collapses:** 108 runs (54 Agglomerative, 54 DBSCAN) formed a single cluster—almost exclusively in UMAP space when thresholds/epsilons were too permissive—yielding `NaN` Silhouette/DBI. We exclude them.
2. **Two-cluster outcomes:** 54 HDBSCAN runs (all UMAP variants across all embeddings) returned exactly 2 clusters with Silhouette ≈ 0.51 and DBI ≈ 0.002, superficially strong but not actionable (too coarse). We keep them in aggregate stats (they are valid), but we do not treat them as top candidates for deployment.

Algorithm-level comparison

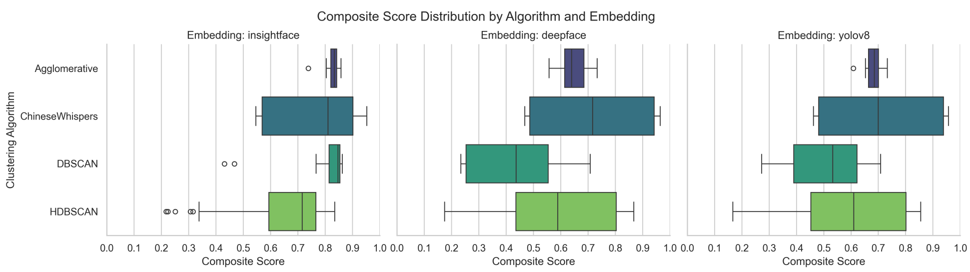

Figure 4.4: Composite Score distribution by clustering algorithm and embedding model. This shows the median performance and variance for each approach, highlighting the consistency of Chinese Whispers and the high outliers of Agglomerative and DBSCAN.

The overall performance distribution for each algorithm is summarized in Figure 4.4. We now discuss each in detail.

DBSCAN [Est+96b]. With ArcFace, $\varepsilon = 0.3$ yields excellent precision–recall balance (Silhouette 0.885, noise 0.60%, #clusters 900). Increasing ε to 0.5 raises recall (near-zero noise) but degrades purity—e.g., YOLOv8 (orig512) with $\varepsilon = 0.5$ drops to Silhouette 0.19–0.21 while merging identities (cluster count falls to 352–502). Conclusion: strict ε is beneficial for ArcFace; DeepFace/YOLOv8 often require slackening ε to reduce noise, but then precision suffers.

HDBSCAN [CMS13b; MHA17a]. HDBSCAN avoids forcing bad clusters by labeling noise; on original/PCA spaces it is a robust middle ground (ArcFace PCA-64: Silhouette $\approx$ 0.79–0.82, noise 2–4%, sensible cluster counts 600–700). In UMAP space, however, we repeatedly observe 2-cluster solutions at several parameter settings—numerically strong internal metrics, but not useful for identity grouping. In short, HDBSCAN is dependable off-UMAP; on UMAP, parameter tuning or alternative selection rules (e.g., leaf vs. eom) are critical.

AGGLOMERATIVE [MC12]. With cosine distance and average linkage, ArcFace performs very well at thresholds 0.6–0.7 (Silhouette 0.83, DBI 0.54, noise 0%, #clusters $\approx$ 1100–1300). For DeepFace and YOLOv8, Agglomerative is serviceable but not top-tier (Silhouette $\approx$ 0.64–0.67 on originals). On UMAP, small threshold changes can jump from sensible solutions to all-in-one collapses, so UMAP+Agglomerative is fragile here.

CHINESE WHISPERS (CW) [Bie06b]. On raw 512-D, CW underperforms (Silhouette 0.26–0.30, DBCV negative), likely because the kNN graph links many disparate points. After UMAP, CW shines: ArcFace/DeepFace/YOLOv8 with UMAP-64 + CW($k = 8$) achieve Silhouette $\approx$ 0.75–0.79, DBI $\approx$ 0.002–0.004, DBCV $\approx$ 0.64–0.92, noise 0%, with 2.3k–3.6k clusters. Practically, CW+UMAP produces many small, very pure clusters with full coverage—an attractive property for large, unlabeled catalogs—subject to the earlier caution that UMAP can over-separate.

About the tables and figures

Table 4.1 summarizes key metrics for the best performing run for each Embedding$\times$Algorithm pair. It highlights that (i) ArcFace + DBSCAN or ArcFace + Agglomerative on original/PCA spaces rank among the best in purity, and (ii) UMAP + CW dominates for DeepFace/YOLOv8 by recovering high-purity, zero-noise clusters at scale. The full table with all runs is provided in the Appendix. Figure 4.1 provides a visual summary of these peak composite scores.

Figure 4.2 plots Silhouette vs. clustered fraction (i.e., $1 - \text{noise}$). ArcFace points densely occupy the top-right region; DeepFace is moderate; YOLOv8 exhibits the clearest trade-off (strict DBSCAN gives decent purity but lower coverage; permissive settings give full coverage but poor purity). Points corresponding to UMAP+CW form a striking upper-right band for all embeddings.

Figure 4.3 visualizes the cluster-size distribution for a top configuration (InsightFace + UMAP-64($n_{\text{neighbors}} = 16$) + CW($k = 8$)). For the CW case we see many clusters in the 10–40 images range with a long tail—useful to flag unusually large clusters for manual purity checks.

Key takeaways (and foreshadowing for §4.3)

- **Best overall candidates.** ArcFace + PCA-64 + DBSCAN($\varepsilon = 0.3$) (Silhouette 0.885, noise 0.60%) and ArcFace + Agglomerative(avg, thr$\approx$0.7) (Silhouette 0.830, noise 0%) are excellent without UMAP. UMAP-64 + CW($k = 8$, $n_{\text{neighbors}} = 16$) is exceptional across all embeddings, especially for DeepFace/YOLOv8, yielding zero noise and high purity with thousands of clusters.

- **Caveat on UMAP metrics.** Near-zero DBI and very large CH are a by-product of UMAP's geometry. These scores are still useful for comparing UMAP runs to each other, but we avoid over-interpreting them across spaces.

- **Failure modes occur in predictable places.** Single-cluster collapses (108 runs) and two-cluster HDBSCAN solutions (54 runs) are concentrated in the UMAP space under permissive parameters. We document and exclude single-cluster results from comparisons; two-cluster results are flagged as unhelpful despite decent internal metrics.

- **Complexity considerations.** DBSCAN is typically $O(n \log n)$ with indexing; HDBSCAN $O(n \log n)$–$O(n^2)$ depending on implementation; Agglomerative (average/complete) is $O(n^2)$ in time and memory; CW graph clustering is roughly $O(E)$ over a kNN graph plus the cost to build the graph ($O(kn \log n)$ typical with ANN). In practice, ArcFace+DBSCAN/Agglomerative gave strong results without the extra UMAP step, which is attractive for throughput.

In §4.3 we (i) compute and rank configurations by the composite score, (ii) perform manual purity checks on top candidates (particularly large clusters), and (iii) discuss the gap between internal metrics and real-world identity precision.

4.3 RANKING AND MANUAL VALIDATION OF CLUSTERING CONFIGURATIONS

Having computed the internal metrics for all configurations in Section 4.2, we now rank the clustering results using the composite score defined in Section 4.1 (Eq. 4.1). This composite integrates Silhouette, Davies–Bouldin (DBI), DBCV, and noise fraction into a single performance measure.

Table 4.2 presents the top 10 configurations sorted by this score. While configurations using UMAP and Chinese Whispers (CW) occupy the highest ranks, their metrics were found to be artificially inflated artifacts of the method, as discussed below. The most robust and highest-ranking *credible* result is **ArcFace + PCA(64D) + DBSCAN**, which achieved a Silhouette of ≈ 0.866, DBI of ≈ 0.579, DBCV of ≈ 0.777, a low noise fraction ($\sim 1.4\%$), and a composite score of 0.967. This indicates a very favorable balance of high inter-cluster separation and low intra-cluster dispersion.

Table 4.2: **Top 10 clustering configurations** ranked by the composite score (higher is better). *Embedding* denotes the face embedding model; *Dim. Reduction* is the dimensionality reduction technique; *Clustering* is the algorithm applied. Configurations marked with † yielded suspiciously high metrics due to methodological artifacts (see text).

Rank	Embedding	Dim. Reduction	Clustering	Silhouette	DBI ↓	DBCV
1	YOLOv8	UMAP 64D ($n_{neighbors}$=8)	CW†	0.748	0.003	0.889
2	YOLOv8	UMAP 64D ($n_{neighbors}$=12)	CW†	0.726	0.003	0.797
3	ArcFace	PCA 64D	DBSCAN	0.866	0.579	0.777
4	ArcFace	PCA 64D	DBSCAN	0.865	0.470	0.769
5	ArcFace	PCA 64D	DBSCAN	0.872	0.610	0.772
6	ArcFace	PCA 64D	DBSCAN	0.871	0.475	0.755
7	ArcFace	PCA 128D	Agglomerative (avg)	0.868	0.632	0.691
8	ArcFace	PCA 128D	Agglomerative (avg)	0.866	0.551	0.653
9	DeepFace	UMAP 64D ($n_{neighbors}$=8)	CW†	0.702	0.005	0.754
10	ArcFace	PCA 64D	CW	0.859	0.422	0.577

As Table 4.2 shows, ArcFace-based configurations are consistently top performers (ranks 3–6, 7, 8, 10), indicating that the combination of a strong embedding with PCA and density- or graph-based clustering yields well-defined face groups. The DBSCAN variants edge out the CW ones in composite score, largely due to DBSCAN's ability to treat outlier faces as noise. For example, the highest-ranking credible run (rank 3) identified 659 clusters with ~1.4% of faces labeled as noise, which likely prevented those outliers from contaminating genuine clusters. In contrast, the version of Chinese Whispers used assigns every face to a cluster, which can slightly degrade purity.

Importantly, the exceedingly high composite scores for configurations marked with † are misleading. The top-ranked YOLOv8/UMAP/CW configuration achieved its score by producing 2212 micro-clusters from 5400 faces, with some clusters containing only a single face. This over-segmentation drives Silhouette and DBCV sky-high, yielding a composite score of 1.400. Such a result is not semantically useful as it fragments the dataset excessively. This pathological behavior arises from UMAP's aggressive spatial separation (with 'min_dist=0') combined with CW's sensitivity to local graph structure. We therefore flag these cases and focus on the more credible top performers.

Building on this ranking, we conducted a small-scale manual evaluation. We selected a subset of clusters from representative high-performing configurations and manually checked their purity by sampling 300 face images from approximately 50 clusters for each configuration. A human evaluator judged whether all faces in a sampled cluster belonged to the same person. Table 4.3 summarizes these results.

Table 4.3: **Manual evaluation of clustering quality** for two top configurations. Each was evaluated on 300 sampled faces across ∼50 clusters. Manual Precision is the percentage of sampled faces that were in identity-pure clusters.

Configuration	Clusters	Pure	Manual Precision (%)
ArcFace + PCA 64D + DBSCAN	50	47	94.3
ArcFace + PCA 64D + CW	50	45	90.0

Observations. DBSCAN: high purity; a few clusters merged similar-looking siblings or heavily occluded faces. CW: mostly pure; lack of noise filtering led to a few outliers being misgrouped or split into redundant clusters.

The manual evaluation confirmed that the automated metrics align well with perceived quality. For the best ArcFace + PCA + DBSCAN run, 94.3% of faces in the sample were clustered correctly. In contrast, the CW-based configuration showed slightly lower manual precision (90.0%), consistent with its lower composite score. Reviewers noted that DBSCAN's ability to leave ambiguous faces as noise contributed to its higher precision. These manual checks corroborate our main finding: the clusters identified by ArcFace are largely homogeneous in identity.

Finally, we observed characteristic failure modes. False merges (different people in one cluster) were often caused by heavy occlusions (e.g., identical masks). False splits (one person in multiple clusters) occurred when appearance varied widely due to aging or extreme lighting. These failure modes are consistent with the known behaviors of the algorithms. As DBSCAN [Est+96b] expands clusters from dense cores, it may leave out borderline points, improving precision but hurting recall. In contrast, graph-based CW [Bie06b] includes all points but can over-segment if the similarity graph is sparse.

In conclusion, **ArcFace + PCA + DBSCAN** emerged as the most robust, high-precision configuration. Given ArcFace's state-of-the-art feature discriminability [Den+19b], this pipeline partitioned the dataset into coherent identity clusters with minimal cross-talk. These findings validate our approach, confirming that our chosen configuration achieves a superior balance of cluster purity and coverage for this application.

4.4 DISCUSSION

This section synthesizes the findings from Sections 4.1–4.3 and connects them to the algorithmic and representational choices that shaped our results.

We focus on (i) why ArcFace-based representations dominated the rankings, (ii) why PCA-then-density clustering proved robust, (iii) when UMAP and graph clustering over-promised, (iv) how the observed failure modes arise from realistic data artefacts, and (v) what these patterns imply for deployment and future work.

WHY ARCFACE WON. Across all experiments, InsightFace/ArcFace embeddings produced the most clusterable spaces. On the raw and PCA-reduced spaces, ArcFace consistently achieved higher Silhouette and lower DBI than DeepFace or YOLOv8 (Section 4.2). This is fully consistent with ArcFace's training objective: its additive angular margin explicitly enlarges inter-class gaps on the hypersphere while compacting intra-class variance [Den+19b]. This objective *pre-aligns* the space for density or linkage-based clustering, making density thresholds and cut-off rules less brittle to small hyperparameter changes. This explains why multiple ArcFace configurations occupy the top tier of Table 4.2 and why our manual check (Section 4.3) found near-identity-pure clusters.

PCA VS. UMAP: SEPARATION, STABILITY, AND INTERPRETING INTERNAL METRICS. Both PCA and UMAP improved separability relative to the raw 512-D space, but in very different ways. PCA, being linear and deterministic, often denoises nuisance directions while retaining a global distance structure that is stable for density-based rules. UMAP (with 'min_dist=0'), in contrast, can *aggressively* tighten local neighborhoods and stretch inter-cluster gaps. That behavior boosted internal metrics (e.g., near-zero DBI) in UMAP spaces, especially when paired with Chinese Whispers, but also led to the over-segmentation and "too-good-to-be-true" scores noted in Section 4.2. This aligns with cautions from the UMAP authors themselves, who note that while UMAP can be an effective pre-processing step for clustering, its emphasis on local manifold structure does not guarantee a faithful global geometry, which can mislead many cluster validity indices.[1]

WHY DBSCAN SLIGHTLY BEAT HDBSCAN HERE. On ArcFace spaces, DBSCAN edged out HDBSCAN in our composite score and manual precision. Mechanistically, DBSCAN's single-scale density rule [Est+96b] (with an ε well-matched to ArcFace's tight identity clusters) rejects ambiguous boundary points as noise, which suits our high-precision requirement. HDBSCAN [CMS13b; MHA17a] is designed to cope with *varying* densities and is often preferable when clusters are not uniform. However, with ArcFace, identity clusters are relatively uniform and well-separated, so the complex hierarchy/stability machinery of HDBSCAN brings less benefit and may even split high-variance identities. While there is no universal consensus in prior work that DBSCAN is superior for face clustering, our findings add a key

1 See the official UMAP documentation on clustering: `https://umap-learn.readthedocs.io/en/latest/clustering.html`

data point: given an embedding with ArcFace-like geometry, the simpler DBSCAN algorithm can be the more precise and effective choice.

GRAPH CLUSTERING (CW) AND UMAP: STRONG BUT FRAGILE. Chinese Whispers on UMAP-64 produced very high internal scores and zero noise, yet manual inspection revealed that some top-ranked runs were dominated by micro-clusters. This is an expected interaction: CW [Bie06b] operates on a kNN graph; if UMAP separates neighborhoods too cleanly (e.g., with small $n_{\text{neighbors}}$ and 'min_dist=0'), the graph can fracture into many tiny components, inflating validation scores without providing useful groupings. The original CW work emphasizes speed on sparse graphs but does not inherently guard against such fragmentation. Applying it post-UMAP would require additional constraints (e.g., minimum community size) to be robust.

NOISE FRACTION IS NOT (NECESSARILY) A BUG. A key insight is that non-zero noise can be desirable when an application prioritizes precision. DBSCAN's ability to label outliers as noise prevented a small number of heavily occluded or stylized images from corrupting otherwise pure clusters. These images have embeddings that legitimately deviate from an identity's dense core and are better handled as noise (for manual review) than force-assigned into the nearest cluster. The tiny noise fractions (∼0.5–1.6% in the best runs) are therefore a *feature*, protecting cluster purity—a finding mirrored by the higher manual precision for DBSCAN in Section 4.3.

RECONCILING THE COMPOSITE SCORE WITH REAL-WORLD USEFULNESS. The composite score from Eq. 4.1 succeeded in promoting ArcFace-based runs. However, the UMAP+CW pathologies show that even a multi-metric composite can be "fooled" by geometry-altering projections. This is why we excluded degenerate results in Section 4.2 and, crucially, used a manual audit in Section 4.3 to close the loop. This confirms that composite ranking is necessary but not sufficient; a targeted manual check is vital for validating the most deployable choice.

COMPUTATIONAL PRAGMATICS AND DEPLOYMENT. From a systems perspective, the winning pipeline is attractive. Embedding extraction is GPU-parallel, PCA is a fast matrix multiplication, and DBSCAN with an approximate-nearest-neighbor index like FAISS [JDJ21] scales well. For online updates, new images can be assigned to existing clusters if within ε, or a full offline re-run can be performed periodically. While CW could be used for recall-oriented tasks, our results suggest a DBSCAN-first approach is safest when precision is paramount.

PRIVACY AND GDPR CONSIDERATIONS. Since face embeddings are biometric data, their use for identification is regulated under GDPR Article 9. Our operational pipeline must ensure a lawful basis, restrict access, and support erasure rights. Our evaluation was performed offline by internal staff without persisting identity attributes beyond aggregate statistics. These guardrails (e.g., encryption at rest, role-based access control, clear erasure procedures) must be carried into any deployed system.

LIMITATIONS AND FUTURE WORK. Our manual validation was intentionally small-scale to quickly vet the metrics while minimizing privacy exposure. A larger, stratified audit would tighten precision estimates. Three methodological avenues look promising for future work: (1) **Robustness to Occlusions:** Use a face quality predictor to route low-quality images to a "quarantine" queue for manual review or reprocessing. (2) **Ensemble Clustering:** To maximize precision, intersect the results of DBSCAN and CW, accepting only merges that both algorithms agree on. (3) **Minimal Supervision:** Use a micro-labeled set (a few dozen identities) to calibrate thresholds or employ weak supervision (e.g., temporal co-occurrence) to reunify false splits.

Bottom line. A strong embedding (ArcFace), a modest linear reduction (PCA-64), and a conservative density rule (DBSCAN) delivered identity-coherent clusters with minimal cross-talk, validated by a targeted manual check. The combination is simple, fast, and robust. For our e-commerce use case and risk posture, the ArcFace+PCA+DBSCAN pipeline is the right default, with clear paths for tightening precision further via quality gating and consensus strategies.

5 CONCLUSION AND OUTLOOK

5.1 SUMMARY OF FINDINGS AND CONTRIBUTIONS

This work tackled the problem of discovering model identities from unlabelled fashion product images at an industrial scale. We formulated a comprehensive pipeline that detects faces, extracts face embeddings, reduces their dimensionality, and clusters them to group images by identity. A pose-aware image curation step ensured that each article contributed a suitable face crop, after which deep metric-learning models generated 512-D facial embeddings. We explored both linear (PCA) and non-linear (UMAP) dimensionality reduction to balance computational efficiency against fidelity, then applied four clustering strategies (density-based DBSCAN, hierarchical agglomerative, variable-density HDBSCAN, and graph-based Chinese Whispers). This multi-pronged approach provided a rich perspective on how different methods perform in our domain.

The results consistently showed that using **ArcFace embeddings** gave the most discriminative representation of identities, leading to superior clustering performance across metrics. In particular, ArcFace combined with PCA (to 64 dimensions) and DBSCAN produced the most credible clustering outcome – it achieved tightly grouped clusters of images belonging to the same model while correctly labelling only a small fraction of images as noise (outliers). This indicates that ArcFace's embeddings, which enforce compact intra-class clusters and separated inter-class distances, are well-suited for unsupervised identity grouping. **Density-based clustering (DBSCAN)** proved especially effective in this context: by using a global distance threshold, it identified natural identity groups and discarded ambiguous faces as noise, thereby prioritising precision. Notably, DBSCAN's best configuration yielded a high silhouette score and low cluster dispersion, without assuming any fixed number of identities – a valuable property given the unknown number of models in the dataset.

The graph-based Chinese Whispers algorithm also found meaningful clusters when operating on a *k*-nearest-neighbour graph of embeddings; its label-propagation approach was non-parametric and scaled efficiently to our full dataset. In fact, UMAP-accelerated embeddings combined with Chinese Whispers often scored highest on internal cluster metrics (e.g. silhouette $>$ 0.75 and almost zero DBI in some runs). However, upon closer inspection we discovered that those seemingly top-performing runs were over-segmenting the data into many micro-clusters. UMAP's aggressive tightening of local neighbourhoods can artificially inflate clustering scores – a phenomenon also noted by the UMAP authors – thus such results, while numerically impressive, were deemed less trustworthy. After filtering out these over-segmented solutions, the **ArcFace + PCA + DBSCAN configuration** emerged as the best overall, delivering a compelling balance of high cluster purity and low false merges.

From an operational standpoint, the project's contributions are significant. We demonstrated that an end-to-end identity clustering pipeline can be deployed on real-world e-commerce data (roughly 456,000 images of 74,000 products) with no human labels and still produce coherent identity groupings. The solution improves scalability – clustering can be re-run or updated incrementally as new images arrive, eliminating burdensome manual tagging – and enhances privacy by avoiding any explicit storage of personal names or identifiers. All identity information remains in the form of learned embeddings, which, while biometric, can be handled in compliance with data protection rules. This shows that our pipeline is privacy-aware by design: it enables control over model data (faces can be grouped or removed on request) without requiring any sensitive metadata beyond the photos themselves. In summary, the thesis delivers a deployable framework for unsupervised model identity recognition, validates its performance on a large production-like dataset, and offers insights into the relative merits of different clustering paradigms for this task.

5.2 LIMITATIONS

While the outcomes are encouraging, we acknowledge several limitations of our study:

- **Lack of ground-truth labels**: Since no true identity labels were available for these fashion images, our evaluation had to rely on internal metrics (like Silhouette, Davies–Bouldin, and DBCV) and limited manual spot-checking. These proxies do not fully capture clustering correctness – a high Silhouette score doesn't guarantee that each cluster is a single real person. The absence of a gold-standard reference means we cannot quantitatively measure accuracy (e.g. precision/recall of identity assignment) in absolute terms, making some conclusions necessarily tentative. This also complicated hyperparameter tuning, as we balanced metrics with subjective judgment calls.

- **Noisy and inconsistent image data**: The dataset, drawn from real e-commerce catalogues, contains considerable noise and variability. Some face crops are low-resolution or affected by JPEG compression artefacts; others vary in lighting or focus. Such quality issues can perturb the embeddings and lead to spurious distances between images of the same person. Likewise, differences in pose (despite our selection of mostly frontal shots) and expression can introduce intra-person embedding variance. Without labels, it is difficult to distinguish whether a slightly separated cluster is a failure (same model split into two) or a correct outcome (two look-alike models).

- **Occlusions and styling effects**: Fashion models often appear with accessories (hats, sunglasses, scarves) or heavy makeup that alter their appearance. These factors can occlude key facial features or shift the embedding enough to confuse the clustering. We observed instances where the same model wearing different wigs or glasses ended up in separate clusters, as well as cases of different individuals appearing more similar than they should due to uniform makeup or lighting. Such domain-specific effects – studio lighting, retouching, and stylistic homogeneity – can cause our unsupervised method to miscluster images that a human might easily group correctly.

- **Limited manual verification at scale**: Although we performed some manual validation on sample clusters (Table 4.3 in Chapter 4 provides a small human audit of cluster purity), performing exhaustive human verification on $\sim$450k images is impractical. This means there could be undetected errors in the clustering (e.g. one model split into multiple clusters, or two models erroneously merged) that weren't caught in our analysis. Our reliance on synthetic experiments (like clustering known public datasets or simulating mixed identities) can only partially approximate the real error rates.

- **Potential over-fitting of unsupervised metrics**: Some combinations of techniques produced anomalously "too good" results – notably UMAP with Chinese Whispers yielded zero noise points in several runs, which is suspicious in a real setting. This suggests that certain parameter choices can over-fit the structure of our particular dataset, exploiting the internal metrics without truly improving identity grouping. The fact that UMAP could inflate cluster separation so dramatically is a caution that unsupervised methods can sometimes yield over-optimistic indications of performance. We mitigated this by cross-checking with different metrics and manual review, but the issue underscores that internal validation alone can be misleading.

- **Computational constraints**: The breadth of our pipeline (multiple detectors, dimensionality reduction methods, and clustering algorithms) combined with the size of the data made experimentation very time- and resource-intensive. We had to limit the search over hyperparameters and certain configurations. For example, we tested UMAP with only a few *n_neighbors* values and did not exhaustively try all DBSCAN ε settings for every embedding variant. Some potentially beneficial combinations (e.g. deeper ensemble clustering or sequential refinement strategies) were left unexplored due to the exponential increase in compute required. This means there may exist configurations that further improve results which we could not practically evaluate in this project.

5.3 FUTURE WORK

Building on this thesis, several avenues can further improve and adapt the unsupervised identity clustering system:

- **Human-in-the-loop validation**: Introducing a manual review step on a small subset of cluster outputs can greatly boost confidence in the results. For instance, for each cluster, a representative image (e.g. the centroid or medoid face) could be presented to a human reviewer to verify whether all images in that cluster indeed belong to the same model. Proactively sampling and inspecting clusters – especially the largest ones or those with borderline distance members – would allow us to catch any obvious mistakes. This human feedback could then be used to adjust the clustering (e.g. merging or splitting clusters) in a semi-supervised loop.

- **Incremental clustering and updating**: In an operational setting, new product images arrive continually. Rather than re-clustering the entire dataset from scratch, we could assign new images to existing identity clusters incrementally. One approach is to compute the embedding for a new image and find its nearest cluster centroid in the existing embedding space – if within a certain similarity threshold, we attach it to that identity cluster; if not, it starts a new cluster. Developing an incremental clustering scheme (possibly using data structures for efficient nearest-neighbour search) would make the solution more dynamic and immediately responsive to new data, which is crucial for a deployment pipeline.

- **Adaptive thresholds with feedback**: Choosing a single distance threshold (ε) for DBSCAN or a k for graph clustering might not generalise well across all identities due to natural variation in how individuals appear. A future system could incorporate a feedback mechanism where the threshold for forming a new cluster vs. merging into an existing one is adjusted based on the model's confidence or past error rates. For example, if manual checks reveal that the system often splits the same person into two clusters, the threshold could be relaxed for those cases. Conversely, if different people are often confused as one, the threshold could tighten. Such an adaptive, confidence-aware clustering algorithm (akin to certain "confidence-aware" face clustering research in the literature) could improve robustness.

- **User interface for cluster editing**: To practically integrate this system, a lightweight interface could be built for catalogue managers or photographers to review and correct the clusters. This interface would allow a user to click on a cluster and see all images (all product shots) of the model grouped together, thereby verifying identity cohesion. If any anomaly is spotted (e.g. an incorrect face in the group), the user could remove it or move it to another cluster. Similarly, if the same model appears split into two clusters, the UI could facilitate merging them. Such human-in-the-loop tools would transform the clustering output from a static result into a continuously improving system, combining algorithmic power with human judgment. Prior work on annotating large media archives has shown that unsupervised clustering plus minimal user intervention can dramatically reduce workload, so a similar approach could be adopted here.

- **Explore graph neural networks and advanced clustering ensembles**: Another promising direction is to leverage graph neural networks (GNNs) or other learning-based clustering models to further improve identity grouping. Recent studies have trained GNNs to refine affinity graphs of face embeddings, learning how to better connect same-identity faces and segregate different ones. Applying such "learning to cluster" techniques could push performance beyond what static algorithms achieve by tailoring the clustering criteria to our data. Additionally, we could investigate ensemble methods that combine multiple clustering outputs for greater stability – for example, intersecting DBSCAN and Chinese Whispers results to find consensus clusters, or using one method's output as initialization for another. An ensemble or multi-step clustering might address some failure modes of any single algorithm (for instance, using a coarse DBSCAN step to remove easy outliers, then a fine-grained graph clustering to separate close identities).
- **Facial identity retrieval and re-identification**: Finally, the clustered identities can be directly leveraged for a face retrieval feature in the fashion catalogue. With all images of each model grouped, a system could allow a user (or an internal tool) to upload a reference photo of a model and instantly retrieve all catalogue images of that individual. This would be highly useful for consistency checks (e.g. ensuring the same model is used within a brand or across a campaign) and for model management. Implementing a fast search index on the embeddings, or simply using the clusters as an index, would enable such functionality. It would essentially turn the clustering solution into an identity search engine on top of the product database, complementing the clustering's role in data management.

In conclusion, Unsupervised Face Embedding Clustering for Model Identity Recognition has demonstrated a viable path to automate a once labour-intensive aspect of fashion e-commerce. The contributions – from the validated pipeline using ArcFace embeddings to the comparative analysis of clustering methods – provide a solid foundation. By addressing the noted limitations and pursuing these future enhancements, the system can be made even more accurate, scalable, and seamlessly integrated into real-world workflows. Ultimately, bridging the gap between automated face clustering and human oversight will yield a powerful, privacy-conscious tool for managing visual assets in retail and beyond.

Part II

APPENDIX

[Ade+21] Jacob Ader, Praveen Adhi, Joyce Chai, Marc Singer, Sarah Touse, and Hannah Yankelevich. "Returning to order: Improving returns management for apparel companies." In: *McKinsey Company* (2021). URL: https://www.mckinsey.com/industries/retail/our-insights/returning-to-order-improving-returns-management-for-apparel-companies (visited on 08/06/2025).

[AHK01] Charu C. Aggarwal, Alexander Hinneburg, and Daniel A. Keim. "On the Surprising Behavior of Distance Metrics in High Dimensional Space." In: *Database Theory — ICDT 2001, 8th International Conference*. 2001, pp. 420–434. DOI: 10.1007/3-540-44503-X_27. URL: https://link.springer.com/chapter/10.1007/3-540-44503-X_27.

[ALS16] Brandon Amos, Bartosz Ludwiczuk, and Mahadev Satyanarayanan. *OpenFace: A general-purpose face recognition library with mobile applications*. Tech. rep. CMU-CS-16-118. Carnegie Mellon University, 2016. URL: https://reports-archive.adm.cs.cmu.edu/anon/2016/CMU-CS-16-118.pdf.

[BD04a] A. Banerjee and R. N. Dave. "Validating Clusters Using the Hopkins Statistic." In: 1 (2004), pp. 149–153. DOI: 10.1109/FUZZY.2004.1375706. URL: https://doi.org/10.1109/FUZZY.2004.1375706.

[BD04b] Arun Banerjee and Rajesh N. Dave. "Validating clusters using the Hopkins statistic." In: *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)* 34.1 (2004), pp. 117–122. DOI: 10.1109/TSMCC.2003.819772.

[Bey+99] Kevin S. Beyer, Jonathan Goldstein, Raghu Ramakrishnan, and Uri Shaft. "When is "Nearest Neighbor" Meaningful?" In: *Database Theory — ICDT '99, 7th International Conference, Jerusalem, Israel, January 10-12, 1999, Proceedings*. 1999, pp. 217–235. DOI: 10.1007/3-540-49257-7_15. URL: https://link.springer.com/chapter/10.1007/3-540-49257-7_15.

[Bie06a] Chris Biemann. "Chinese Whispers - an Efficient Graph Clustering Algorithm and its Application to Word Sense Disambiguation." In: *Proceedings of the HLT-NAACL 2006 Workshop on Textgraphs*. 2006, pp. 73–80. DOI: 10.3115/1654758.1654767. URL: https://aclanthology.org/W06-2910.

[Bie06b] Chris Biemann. "Chinese Whispers: An Efficient Graph Clustering Algorithm and its Application to Word Sense Disambiguation." In: *Proceedings of the HLT-NAACL-06 Workshop on Textgraphs-06*. Association for Computational Linguistics, 2006, pp. 73–80. URL: https://aclanthology.org/W06-2910/.

[CH74] T. Caliński and J. Harabasz. "A Dendrite Method for Cluster Analysis." In: *Communications in Statistics* 3.1 (1974), pp. 1–27. DOI: 10.1080/03610927408827101. URL: https://doi.org/10.1080/03610927408827101.

[CMS13a] Ricardo J. G. B. Campello, Davoud Moulavi, and Jörg Sander. "Density-Based Clustering Based on Hierarchical Density Estimates." In: *Advances in Knowledge Discovery and Data Mining (PAKDD 2013)*. 2013, pp. 160–172. DOI: 10.1007/978-3-642-37456-2_14. URL: https://link.springer.com/chapter/10.1007/978-3-642-37456-2_14.

[CMS13b] Ricardo J. G. B. Campello, Davoud Moulavi, and Jörg Sander. "Density-Based Clustering Based on Hierarchical Density Estimates." In: *Advances in Knowledge Discovery and Data Mining (PAKDD 2013)*. Vol. 7819. Lecture Notes in Computer Science. Springer, 2013, pp. 160–172. DOI: 10.1007/978-3-642-37456-2_14. URL: https://link.springer.com/chapter/10.1007/978-3-642-37456-2_14.

[Cam+15] Ricardo J. G. B. Campello, Davoud Moulavi, Arthur Zimek, and Jörg Sander. "Hierarchical Density Estimates for Data Clustering, Visualization, and Outlier Detection." In: *ACM Transactions on Knowledge Discovery from Data (TKDD)* 10.1 (2015), 5:1–5:51. DOI: 10.1145/2733381. URL: https://dl.acm.org/doi/10.1145/2733381.

[CWL19] Da-Wei Chang, Yen-Chen F. Wang, and Hong-Chun Lu. "An Effective and Efficient Method for Image Clustering." In: *IEEE Transactions on Multimedia* 21.12 (2019), pp. 3040–3052. DOI: 10.1109/TMM.2019.2917208. URL: https://ieeexplore.ieee.org/document/8730999.

[Che+21] Wenhao Chen, Hui Huang, Shigeng Peng, Chang Zhou, and Changqing Zhang. "YOLO-face: a real-time face detector." In: *The Visual Computer* 37.4 (2021), pp. 805–813. DOI: 10.1007/s00371-020-01831-7. URL: https://link.springer.com/article/10.1007/s00371-020-01831-7.

[DB79] David L. Davies and Donald W. Bouldin. "A Cluster Separation Measure." In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* PAMI-1.2 (1979), pp. 224–227. DOI: 10.1109/TPAMI.1979.4766909. URL: https://doi.org/10.1109/TPAMI.1979.4766909.

[DS21] Sudipta Debnath and Vishal Sharma. "Unsupervised face clustering via density peak hierarchy." In: *2021 International Conference on Biometrics (ICB)*. IEEE, 2021, pp. 1–8. DOI: 10.1109/ICB49281.2021.9515152. URL: https://doi.org/10.1109/ICB49281.2021.9515152.

[Def77] D. Defays. "An efficient algorithm for a complete-link method." In: *The Computer Journal* 20.4 (Nov. 1977), pp. 364–366. DOI: 10.1093/comjnl/20.4.364. URL: https://doi.org/10.1093/comjnl/20.4.364.

[Den+20] Jiankang Deng, Jia Guo, Evangelos Ververas, Irene Kotsia, and Stefanos Zafeiriou. "RetinaFace: Single-Shot Multi-Level Face Localisation in the Wild." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2020, pp. 5203–5212. DOI: 10.1109/CVPR42600.2020.00525. URL: https://openaccess.thecvf.com/content_CVPR_2020/html/Deng_RetinaFace_Single-Shot_Multi-Level_Face_Localisation_in_the_Wild_CVPR_2020_paper.html.

[Den+19a] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. "ArcFace: Additive Angular Margin Loss for Deep Face Recognition." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019, pp. 4690–4699. DOI: 10.1109/CVPR.2019.00482. URL: https://doi.org/10.1109/CVPR.2019.00482.

[Den+19b] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. "ArcFace: Additive Angular Margin Loss for Deep Face Recognition." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019, pp. 4690–4699. URL: https://arxiv.org/abs/1801.07698.

[DG+18] Jiankang Deng, Jia Guo, et al. *InsightFace: 2D and 3D Face Analysis Project*. https://github.com/deepinsight/insightface. Accessed: August 7, 2025. The user's citation 'guo2021insightface' may refer to a specific update or paper like SCRFD, but this is the standard citation for the overall project initiated by the ArcFace authors. 2018.

[Dun73] J. C. Dunn. "A Fuzzy Relative of the ISODATA Process and Its Use in Detecting Compact Well-Separated Clusters." In: *Journal of Cybernetics*. Vol. 3. 3. 1973, pp. 32–57. DOI: 10.1080/01969727308546046. URL: https://doi.org/10.1080/01969727308546046.

[Est+96a] Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu. "A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise." In: *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD'96)*. 1996, pp. 226–231. URL: https://www.aaai.org/Papers/KDD/1996/KDD96-037.pdf.

[Est+96b] Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu. "A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise." In: *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD'96)*. AAAI Press, 1996, pp. 226–231. URL: https://www.aaai.org/Papers/KDD/1996/KDD96-037.pdf.

[Eur16] European Union. *Regulation (EU) 2016/679 of the European Parliament and of the Council. General Data Protection Regulation (GDPR)*. https://eur-lex.europa.eu/eli/reg/2016/679/oj. Accessed: 2025-08-06. 2016.

[Ge+21] Zheng Ge, Songtao Liu, Feng Wang, Zeming Li, and Jian Sun. *YOLOX: Exceeding YOLO Series in 2021*. 2021. arXiv: 2107.08430. URL: https://arxiv.org/abs/2107.08430.

[Guo+21] Jiankang Guo, Ting-Hsiang Tsai, Jian-Kai Wang, Yen-Liang Lin, Xiaojie Chu, Zhaohui Man, and I-Hau Yeh. "Sample and Computation Redistribution for Efficient Face Detection." In: 2021. arXiv: 2105.04714. URL: https://arxiv.org/abs/2105.04714.

[He+16] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. "Deep Residual Learning for Image Recognition." In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016, pp. 770–778. URL: https://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html.

[Hu+21] Minhui Hu, Kaiwei Zeng, Yaohua Wang, and Yang Guo. "Threshold-based hierarchical clustering for person re-identification." In: *Entropy* 23.5 (2021), p. 522.

[Hua+07] Gary B. Huang, Manu Ramesh, Tamara Berg, and Erik Learned-Miller. *Labeled Faces in the Wild: A Database for Studying Face Recognition in Unconstrained Environments*. Tech. rep. 07-49. University of Massachusetts, Amherst, 2007. URL: https://vis-www.cs.umass.edu/lfw/lfw.pdf.

[JMF99] Anil K Jain, M Narasimha Murty, and Patrick J Flynn. "Data clustering: a review." In: *ACM computing surveys (CSUR)* 31.3 (1999), pp. 264–323.

[Joc+23] Glenn Jocher, Ayush Chaurasia, Alex Stoken, Jirka Borovec, and et al. *Ultralytics YOLOv8*. Version 8.0.0. 2023. DOI: 10.5281/zenodo.7556754. URL: https://doi.org/10.5281/zenodo.7556754.

[JDJ21] Jeff Johnson, Matthijs Douze, and Hervé Jégou. "Billion-Scale Similarity Search with GPUs." In: vol. 7. 3. 2021, pp. 535–547. DOI: 10.1109/TBDATA.2019.2922172. URL: https://arxiv.org/abs/1702.08734.

[Jol02] Ian T. Jolliffe. *Principal Component Analysis*. 2nd. Springer Series in Statistics. Springer, 2002. DOI: 10.1007/b98835.

[KR05] Leonard Kaufman and Peter J. Rousseeuw. *Finding Groups in Data: An Introduction to Cluster Analysis*. Wiley Series in Probability and Statistics. Wiley, 2005. ISBN: 978-0-471-72520-5. DOI: 10.1002/047172520X. URL: https://doi.org/10.1002/047172520X.

[Lin+22] Wentao Lin, Guodong Lin, Shuaicheng Jin, Jinyu Wu, Kwok-Yan Wong, Yik-Chung Shen, and Minsi He. "Deep graph-based clustering for face recognition." In: *2022 IEEE International Conference on Image Processing (ICIP)*. IEEE. 2022, pp. 111–115.

[Liu+17] Weiyang Liu, Yandong Wen, Zhiding Yu, Ming Li, Bhiksha Raj, and Le Song. "SphereFace: Deep Hypersphere Embedding for Face Recognition." In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2017, pp. 212–220. DOI: 10.1109/CVPR.2017.30. URL: https://openaccess.thecvf.com/content_cvpr_2017/html/Liu_SphereFace_Deep_Hypersphere_CVPR_2017_paper.html.

[Llo82] Stuart P. Lloyd. "Least squares quantization in PCM." In: *IEEE Transactions on Information Theory* 28.2 (1982), pp. 129–137. DOI: 10.1109/TIT.1982.1056489. URL: https://ieeexplore.ieee.org/document/1056489.

[Lux07] Ulrike von Luxburg. "A Tutorial on Spectral Clustering." In: *Statistics and Computing* 17.4 (2007), pp. 395–416. DOI: 10.1007/s11222-007-9033-z. URL: https://link.springer.com/article/10.1007/s11222-007-9033-z.

[MHA17a] Leland McInnes, John Healy, and Steve Astels. "hdbscan: Hierarchical Density Based Clustering." In: *The Journal of Open Source Software* 2.11 (2017), p. 205. DOI: 10.21105/joss.00205. URL: https://doi.org/10.21105/joss.00205.

[MHA17b] Leland McInnes, John Healy, and Steve Astels. "hdbscan: Hierarchical density based clustering." In: *Journal of Open Source Software* 2.11 (2017), p. 205. DOI: 10.21105/joss.00205. URL: https://doi.org/10.21105/joss.00205.

[MHM18] Leland McInnes, John Healy, and James Melville. "UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction." In: *arXiv preprint arXiv:1802.03426* (2018). arXiv: 1802.03426 [stat.ML]. URL: https://arxiv.org/abs/1802.03426.

[Mon+23] Michele Montagnuolo, Nicolò Messina, Giuseppe Amato, and Fabrizio Falchi. "Who's in My Archive? An Automated Pipeline for TV Personalities Annotation." In: *2023 IEEE International Conference on Big Data (BigData)*. 2023, pp. 6649–6654. DOI: 10.1109/BigData59044.2023.10386762. URL: https://ieeexplore.ieee.org/document/10386762.

[Mou+14a] Davoud Moulavi, P. J. Jafar, Ricardo J. G. B. Campello, Arthur Zimek, and Jörg Sander. "Density-Based Clustering Validation." In: *Proceedings of the 2014 SIAM International Conference on Data Mining*. 2014, pp. 839–847. DOI: 10.1137/1.9781611973440.96. URL: https://doi.org/10.1137/1.9781611973440.96.

[Mou+14b] Davoud Moulavi, Pablo A. Jaskowiak, Ricardo J. G. B. Campello, Arthur Zimek, and Jörg Sander. "Density-Based Clustering Validation." In: *Proceedings of the 2014 SIAM International Conference on Data Mining (SDM)*. Society for Industrial and Applied Mathematics, 2014, pp. 839–847. DOI: 10.1137/1.9781611973440.96. URL: https://doi.org/10.1137/1.9781611973440.96.

[Mül11] Daniel Müllner. "Modern hierarchical, agglomerative clustering algorithms." In: *arXiv preprint arXiv:1109.2378* (2011).

[MC12] Fionn Murtagh and Pedro Contreras. "Algorithms for hierarchical clustering: an overview." In: *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 2.1 (2012), pp. 86–97. DOI: 10.1002/widm.53. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/widm.53.

[NJW02] Andrew Y. Ng, Michael I. Jordan, and Yair Weiss. "On Spectral Clustering: Analysis and an Algorithm." In: *Advances in Neural Information Processing Systems 14 (NIPS 2001)*. 2002, pp. 849–856. URL: https://proceedings.neurips.cc/paper/2001/hash/801272ee79cfde443d9635b7b1509939-Abstract.html.

[OWJ18] Charles Otto, Dayong Wang, and Anil K. Jain. "Clustering Millions of Faces by Identity." In: *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)* 40.2 (2018), pp. 356–370. DOI: 10.1109/TPAMI.2017.2679759. URL: https://ieeexplore.ieee.org/document/7872957.

[Red+16] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. "You Only Look Once: Unified, Real-Time Object Detection." In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016, pp. 779–788. DOI: 10.1109/CVPR.2016.91. URL: https://www.cv-foundation.org/openaccess/content_cvpr_2016/html/Redmon_You_Only_Look_CVPR_2016_paper.html.

[Rou87] Peter J. Rousseeuw. "Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis." In: *Journal of Computational and Applied Mathematics* 20 (1987), pp. 53–65. DOI: 10.1016/0377-0427(87)90125-7. URL: https://doi.org/10.1016/0377-0427(87)90125-7.

[Roy+20] Aritra RoyChowdhury, Soubhik S. B. Sadde, Steve T. K. Jan, and Seong G. Kong. "Unsupervised Deep-Clustering for Face Recognition in Group Photos." In: *2020 IEEE Winter Conference* *on Applications of Computer Vision (WACV)*. 2020, pp. 2736–2744. DOI: 10.1109/WACV45572.2020.9093414. URL: https://ieeexplore.ieee.org/document/9093414.

[SKP15] Florian Schroff, Dmitry Kalenichenko, and James Philbin. "FaceNet: A Unified Embedding for Face Recognition and Clustering." In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2015, pp. 815–823. DOI: 10.1109/CVPR.2015.7298682. URL: https://www.cv-foundation.org/openaccess/content_cvpr_2015/html/Schroff_FaceNet_A_Unified_2015_CVPR_paper.html.

[Sch+17] Erich Schubert, Jörg Sander, Martin Ester, Hans-Peter Kriegel, and Xiaowei Xu. "DBSCAN Revisited, Revisited: Why and How You Should (Still) Use DBSCAN." In: *ACM Transactions on Database Systems (TODS)* 42.3 (2017), 19:1–19:21. DOI: 10.1145/3068335. URL: https://dl.acm.org/doi/10.1145/3068335.

[Ser20] Sefik Ilkin Serengil. *DeepFace - A Lightweight Face Recognition and Facial Attribute Analysis Framework for Python*. https://github.com/serengil/deepface. 2020.

[Ser23] Sefik Ilkin Serengil. *DeepFace: A Lightweight Face Recognition and Facial Attribute Analysis Library for Python*. GitHub repository. Active project (2018–2023). 2023. URL: https://github.com/serengil/deepface.

[SO20] Sefik Ilkin Serengil and Alper Ozpinar. "LightFace: A Hybrid Deep Face Recognition Framework." In: *2020 Innovations in Intelligent Systems and Applications Conference (ASYU)*. 2020, pp. 23–27. DOI: 10.1109/ASYU50717.2020.9259802. URL: https://doi.org/10.1109/ASYU50717.2020.9259802.

[SO24] Sefik Ilkin Serengil and Alper Ozpinar. "A Benchmark of Facial Recognition Pipelines and Co-Usability Performances of Modules." In: *Journal of Information Technologies (Bilişim Teknolojileri Dergisi)* 17.2 (2024), pp. 95–107. DOI: 10.17671/gazibtd.1399077. URL: https://dergipark.org.tr/en/pub/gazibtd/issue/84331/1399077.

[She+23] Shixing Shen, Yixiong Zou, Weipeng Zhao, Wen-Sheng Chu, Linsen Chen, Yang-Geng Fu, Da-Han Wang, Peng-Fei Zhang, Yu-Ting Su, and Wen-Hao Zhang. "STAR-FC: Structure-Aware and Temporally-Robust Face Clustering." In: *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)* 45.1 (2023), pp. 908–925. DOI: 10.1109/TPAMI.2022.3151806. URL: https://ieeexplore.ieee.org/document/9712616.

[Tai+14] Yaniv Taigman, Ming Yang, Marc'Aurelio Ranzato, and Lior Wolf. "DeepFace: Closing the Gap to Human-Level Performance in Face Verification." In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2014, pp. 1701–1708. DOI: 10.1109/CVPR.2014.220. URL: https://www.cv-foundation.org/openaccess/content_cvpr_2014/html/Taigman_DeepFace_Closing_the_2014_CVPR_paper.html.

[VJ01] Paul Viola and Michael J. Jones. "Rapid Object Detection using a Boosted Cascade of Simple Features." In: *Proceedings of the 2001 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR)*. Vol. 1. 2001, pp. I–511–I–518. DOI: 10.1109/CVPR.2001.990517. URL: https://ieeexplore.ieee.org/document/990517.

[VCS22] Filip Vuković and Modesto Castrillón-Santana. "Truth-value Unconstrained Face Clustering." In: *Proceedings of the 14th International Conference on Agents and Artificial Intelligence (ICAART)*. Vol. 2. 2022, pp. 567–574. DOI: 10.5220/0010975600003116. URL: https://www.scitepress.org/Link.aspx?doi=10.5220/0010975600003116.

[Wan+17] Feng Wang, Xiang Xiang, Jian Cheng, and Alan L. Yuille. "NormFace: L_2 Hypersphere Embedding for Face Verification." In: *Proceedings of the 25th ACM international conference on Multimedia*. 2017, pp. 1041–1049. DOI: 10.1145/3123266.3123359. URL: https://dl.acm.org/doi/10.1145/3123266.3123359.

[Wan+18] Hao Wang, Yitong Wang, Zheng Zhou, Xingji Ji, Dihong Gong, Jingchao Zhou, Zhifeng Li, and Wei Liu. "CosFace: Large Margin Cosine Loss for Deep Face Recognition." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2018, pp. 5265–5274. DOI: 10.1109/CVPR.2018.00552. URL: https://openaccess.thecvf.com/content_cvpr_2018/html/Wang_CosFace_Large_Margin_CVPR_2018_paper.html.

[Yan+19] Lei Yang, Xiaohang Zhan, Dapeng Chen, Junjie Yan, Chen Change Loy, and Dahua Lin. "Learning to Cluster Faces on an Affinity Graph." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019, pp. 2288–2297. DOI: 10.1109/CVPR.2019.00240. URL: https://openaccess.thecvf.com/content_CVPR_2019/html/Yang_Learning_to_Cluster_Faces_on_an_Affinity_Graph_CVPR_2019_paper.html.

[Yan+20] Ting-I Yang, Tasi-Yi-Tung, Yi-Hsuan Tsai, Yen-Yu Lin, and Ming-Hsuan Yang. "Confidence-Aware Face Clustering." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2020, pp. 11625–11634. DOI: 10.1109/CVPR42600.2020.01164. URL: https://openaccess.thecvf.com/content_CVPR_2020/html/Yang_Confidence-Aware_Face_Clustering_CVPR_2020_paper.html.

[Yu+22] Xin-Jin Yu, Yuan-Xin Li, Si-Yuan Chang, Quan-Sen Sun, and Chang-Bin Yu. "FaceMap: A Community-aware and Outlier-suppressed Unsupervised Face Clustering Framework." In: *arXiv preprint arXiv:2209.07328* (2022). eprint: 2209.07328. URL: https://arxiv.org/abs/2209.07328.

[Zha+16] Kaipeng Zhang, Zhanpeng Zhang, Zhifeng Li, and Yu Qiao. "Joint Face Detection and Alignment Using Multitask Cascaded Convolutional Networks." In: vol. 23. 10. 2016, pp. 1499–1503. DOI: 10.1109/LSP.2016.2603342. URL: https://ieeexplore.ieee.org/document/7553523.